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RARE EVENT ALGORITHMS
USING AI BASED WEATHER
FORECASTING.

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de l'Environnement, UMR 8212 CEA-
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¹Promis, le texte dans mes figures sera assez grand pour être lu du premier coup dans le futur 😊.



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INTRODUCTION

Heatwaves are considered one of the worst consequences of anthropogenic climate change (IPCC, 2023a; Russo et al., 2015) and one to which society will have to adapt as their intensity and duration is bound to grow (Figure 1). In the last 3 decades, significant heatwaves have occurred such as in 2003 and 2019 in Europe, 2010 in Russia, and more recently 2021 in Canada (with temperatures reaching 49.6°C in Lytton, beating previous records by 5°C). In 2025, there also was eastern Mediterranean heatwaves with Portugal and Spain setting new records of 46.6°C and 46°C respectively. According to IPCC's sixth report (IPCC, 2023b), a global warming of 1.5°C will be the cause of 30 000 death per summer in Europe, mostly impacting those most vulnerable. A 3°C global warming would cause unavoidable risks on health and therefore an augmentation of climate inequalities with communities who have historically contributed the least to current climate change being disproportionately affected. Europe is also the fastest warming continent (Copernicus Climate Change Service (C3S) and World Meteorological Organization (WMO), 2025) and the 2015-2025 years were the hottest 11 years on record (State of the Global Climate 2025, 2025). In a broader point of view, it is essential to better understand climate extremes as high-impact weather and climate events cause a significant amount of deaths and cost billions every year.

As a final year student at INSA Rennes in the applied mathematics department, I had the opportunity to do my masters thesis/internship at the Laboratory for Climate and Environmental Sciences (LSCE) based in Gif-sur-Yvette near Paris. This internship followed my M1 internship I completed in the same place and offered me the chance to work on topics related to climate to which I am sensitive.

The research laboratory where I completed my internship is located at the Orme des Merisiers site of the Atomic Energy and Renewable Energies Commission (CEA) in Saclay. The LSCE is a Joint Research Unit (UMR) between the CEA, the National Centre for Scientific Research (CNRS), and the University of Versailles Saint-Quentin (UVSQ), and is part of the Pierre Simon Laplace Institute (IPSL) and Paris-Saclay University. I will not dwell here on a comprehensive overview of the IPSL, the CEA, or the UVSQ, but rather focus on the LSCE. The LSCE brings together over 350 researchers, engineers, technicians, and administrative staff, including more than 150 permanent staff members from its three parent institutions, along with several dozen doctoral students. The LSCE is organized into three scientific themes. Each theme consists of 4 to 6 teams, for a total of 16 scientific teams. The research themes are as follows: "Archives and Tracers," which aims to understand the dynamics and natural variability of the climate across different temporal and spatial scales; "Biogeochemical Cycles and Transfers in the Environment," which aims to characterize and quantify the biogeochemical cycles of certain compounds (particularly greenhouse gases) and the transfer of matter within and between various environmental compartments (air-water, air-land surface, and water-land surface interfaces); and finally, "Climate and Cycles – Modeling of their variabilities and interactions," the theme within which I completed my internship. The various teams within this latter theme focus respectively on the variability and interactions among different components of the Earth system (MERMAID), the integrated dynamics of the climate across multiple time scales (CLIM), numerical modeling infrastructure (CALCULS), and statistical methods for studying variability, extremes, or downscaling (ESTIMR). The team I joined is the ESTIMR team. The ESTIMR team's main objectives are the understanding and modeling of climatic and environmental variability at various spatial scales—ranging from very large structures associated with atmospheric dynamics to highly local phenomena— and temporal scales for the study of past climates, present processes, and future evolutions. Within this broad framework, one of the strengths of the ESTIMR team lies in the use and development of state-of-the-art statistical models tailored to climate issues, facilitated by sustained multidisciplinary interaction encompassing climatology, modeling, physics, statistics, and artificial intelligence. The laboratory provides a highly dynamic and welcoming environment.

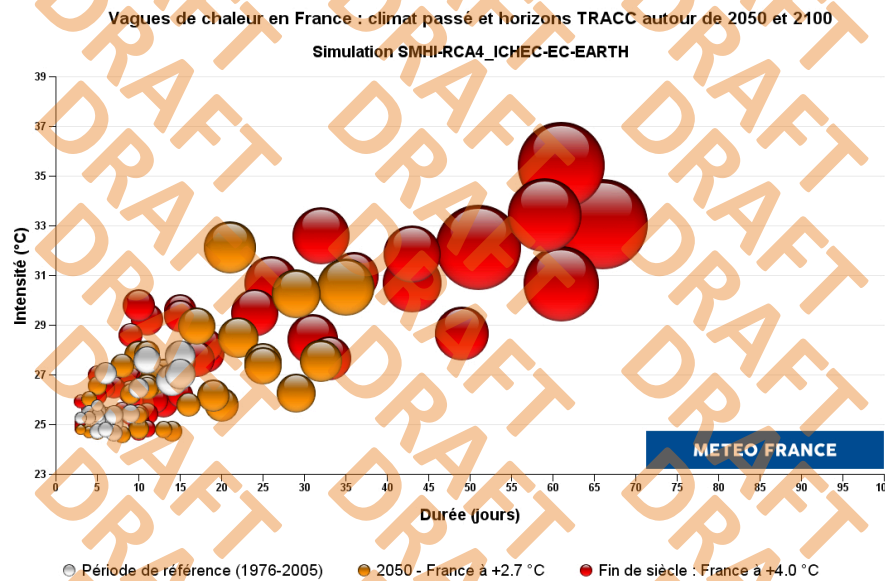


Figure 1 : Heatwaves on the refence period and on TRACC's horizons² of a +2.7°C et +4.0°C France, simulated by the climate model SMHI-RCA4_ICHEC-EC-EARTH. The x-axis represents the duration of the episode. The y-axis represents its intensity, corresponding to the average temperature of the hottest day of the episode. The size of the bubbles corresponds to the severity, which indicates the amount of heat experienced throughout the entire episode. The colors of the bubbles correspond to past or future periods.

During this internship, I worked mostly by myself with instructions given by my supervisors Pascal Yiou and Soulihanh Thao. I also worked during the first month of my internship every Thursday with Jeanne Vedel, a student from ENS Paris-Saclay doing a research project with Pascal Yiou as part of her M2 and with Maeve Mayer, a PhD student. The common denominator for all of us was the better understanding of heatwaves, their associated physical processes and the maximal reachable temperatures. In my case, this is done with artificial intelligence (AI) based models.

Indeed, AI based weather forecasting (ECMWF's AIFS-Single (Lang et al., 2024), AIFS-Ens (Lang et al., 2026) and others, Google (Alet et al., 2025; Lam et al., 2023; Price et al., 2025), Huawei (Bi et al., 2023), NVIDIA (Kossaihi et al., 2026)) is becoming the new paradigm for fast and accurate short and medium range forecasting. These models are (once trained) up to 10000 times (Bi et al., 2023) faster than classical NWP (Numerical Weather Prediction) and can outperform them in some situations (Lang et al., 2024). These encouraging results motivate the use of said models in the context of rare events algorithms ((Fischer et al., 2023), (Ragone et al., 2018), (Bloin-Wibe et al., 2025)), whose goal in climate sciences is to simulate extreme events and estimate return periods or levels. Up to know, these algorithms have been using physical models but climate and weather simulations are notoriously expensive to run. Even with the use of rare event algorithms, the sheer amount of compute power and time to run the different trajectories is significative. Research has been made on using AI to improve these algorithms via hybrid frameworks (physical model + AI) (Lancelin et al., 2026) but no one tried yet a fully AI framework. Using AI based models could help reduce both the compute time and power needed to make studies on extreme events, giving possibilities such as near real time analysis or better results from running more trajectories at a lower cost than with NWP. Nonetheless, there are some remaining questions with regard to AI based weather forecasting models. For example, regarding their chaotic behaviour, their

²<https://www.drias-climat.fr/accompagnement/sections/417>

physical consistency and whether they are able to generalise to unseen future climates (warmer climate for example) as they were trained on past reanalysis data such as ERA5 (Hersbach et al., 2020). They have also been shown to perform poorly in the context of extremes events (Zhang et al., 2025), especially for unseen events. This justifies a more in depth study of those models before attempting to perform rare event algorithms.

The question we are trying to answer to is the following: are AI based weather forecasting models usable in the context of rare events algorithms?

To address this question, this report is structured as follows: In the first section (Section 1), I will briefly present the data used and computing environment I worked in. Then in the second section (Section 2), I will present theoretical background and methodology needed for the understanding of this work and methods used. Covering chaos theory, Lyapunov exponents, the different AIFS versions, key physics principles such as a general explanation of the processes behind heatwaves, the moist static energy, a eulerian decomposition of the temperature, and rare event sampling algorithms, with ensemble boosting and our new iterative boosting. In the third section (Section 3), I start by evaluating the statistical and physical consistency of the AIFS models (summarized in this part, with detailed result in Sup D), I will present the main experimental results, with a case study of the 2026 Heatwave, and a real-world test-case for iterative boosting on the maximal summer temperature. Finally, the key findings, limitations, and future perspectives for AI-driven rare event sampling are addressed in the Discussion and Conclusion sections.

1 DATA AND ENVIRONMENT

For all the coding, I will use python with xarray (Hoyer & Hamman, 2017), pandas (The pandas development team, 2026), matplotlib (Hunter, 2007), dask (Dask Development Team, 2016), numpy (Harris et al., 2020), cartopy (Met Office, 2010), anemol (Lang et al., 2024) and torch (Paszke et al., 2019) on two High Performance Computers (HPC), one (spirit cluster) for the preprocessing and analysis and another (Jean-Zay cluster) for anything that needs a GPU. The GPUs used are NVIDIA A100 with 80gb of VRAM.

For the data, I will mostly work with reanalysis data from the European Centre (ECMWF) ERA5 (Hersbach et al., 2020), which are the most comprehensive and accurate data for weather and the recent past climate. Reanalyses are a mix of observations and short-term past forecasts using modern meteorological models. Consequently, this data is “gapless”, meaning the fields are continuous in both time and space, with a horizontal resolution of 0.28° and hourly timesteps. To understand climate change and extremes, it is essential to have observations spanning as far back as possible. However, a model based solely on past observations would be incomplete and inaccurate. Observations have always been non-uniformly distributed; they can be biased, and even in the satellite era, they are insufficient to provide a complete representation of the state of the climate at a given time t . Reanalyses mitigate these issues. In practical terms, reanalyses combine short-term past forecasts with observations through data assimilation (Bouttier & Courtier, 2002). The objective is to produce daily forecasts, using the current state of the system as initial conditions each time, and then assimilating the available real-world observations.

Recurrent variable used are recapped in Table 1.

2 THEORETICAL BACKGROUND AND METHODOLOGY

This chapter will aim at introducing all the theoretical notions and methods used in this report.

Table 1 : Variables used throughout the study.

Variable	Unit	Description
T_s (or $2t$)	K	Surface (2-meter) air temperature.
T	K	Air temperature in pressure coordinates.
z_s	m	Orography (surface elevation).
z	m	Geopotential height in pressure coordinates. Corresponds to the altitude at which l hPa pressure is reached with respect to the sea level ³ . This helps identifying levels of constant pressure.
q_s	kg/kg	Surface (2-meter) specific humidity, mass of water vapour per kilogram of moist air.
u, v	m/s	Zonal (eastward) and meridional (northward) components of wind in pressure coordinates.
V_h	m/s	Horizontal wind vector (u and v).
ω	Pa/s	Vertical wind velocity in pressure coordinates.
p	Pa or hPa	Atmospheric pressure.
α	m ³ /kg	Specific volume of air ($R\frac{T}{p}$).
e_{sat}	hPa	Saturation vapour pressure.
q_{sat}	kg/kg	Saturation specific humidity.
MSE_s	J/kg	Moist static energy at the surface.
MSE_{500}^*	J/kg	Saturation moist static energy at 500 hPa.
Q_{diab}	K/s	Diabatic heating rate.

2.1 The climate system

Throughout this report, I will be working with a system which I will call the climate system. The sixth IPCC report's glossary (IPCC, 2023b) defines it as “The global system consisting of five major components: the atmosphere, the hydrosphere, the cryosphere, the lithosphere and the biosphere and the interactions between them. The climate system changes in time under the influence of its own internal dynamics and because of external⁴ forcings such as volcanic eruptions, solar variations, orbital forcing, and anthropogenic forcings such as the changing composition of the atmosphere and land-use change”. Essentially, it is the system representing all the components of our earth and their interactions. It is uncertain for two reasons, the first being the use of PDEs and parametrizations⁵ to represent physical laws and the second their numerical solving. They both are approximation used to model and represent the climate system. This makes the performances of models conditioned to our knowledge and representation of the system, with some epistemic uncertainties (Mansfield & Christensen, 2026). Generally, this system is modeled as

$$\frac{dX}{dt} = b(X(t), t, \theta(t)) \quad (1)$$

where $X(t) \in \Omega \subset \mathbb{R}^d$ represents the state vector of the system that includes all the variables that

³Pressure decreases with altitude

⁴Natural

⁵Parametrizations are used for processes often of smaller scale than the model grid. They need to be represented in terms of mean gridbox values, i.e. to be parametrized. Examples are radiation or convection schemes such as precipitation.

represent it at time t , b being the evolution equation⁶ in time, and $\theta(t)$ the forcing applied on the system (greenhouse gases emissions for example). This is what I will be referring to when talking about climate system, assuming AIFS (Lang et al., 2024) learnt it. Note that, we will use interchangeably the terms numerical weather predictions (NWP) and climate models to talk about actual physical models. The difference between these models lies only in their use cases but they model the same system, the first are used for short times scales (up to ~ 15 days) while the latters are used mostly for long scale simulations (decades to centuries).

2.2 Chaos and Lyapunov exponents

While the early works on dynamical systems of Poincaré (Poincaré, 1881) showed the existence of unstable solution to differential equations, the theory of chaos and butterfly effect as we know it emerges from the studies of mathematicians (Lyapunov, 1892) but mostly Lorenz's works (Lorenz, 1963). Lorenz's work showed that so called chaotical system have intriguing properties:

1. The existence of aperiodic but bounded solutions strongly sensitive to initial conditions.
2. The convergence towards complex topological objects in the phase space named strange attractors which is neither a fixed point nor a periodic behaviour.

In other words, these systems exhibit chaotic properties but still show some form of order on longer timescales.

As an example, consider the following system with $\sigma = 10$, $r = 28$ and $b = 8/3$.

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}\tag{2}$$

The system (2) with the parameters we set is a famous chaotic system as seen in Figure 2.

Consider now a dynamical system $\frac{dX}{dt} = f(X(t), t)$. Let $\{X(t)\}_{t \geq 0}$ be a trajectory starting from $X(0) = X_0$ and $\{\tilde{X}(t)\}_{t \geq 0}$ a trajectory starting from perturbed initial conditions $\tilde{X}(0) = \tilde{X}_0$ such that $\delta_0 = X(0) - \tilde{X}(0)$ the perturbation vector at time 0 is small. We write

$$|\delta_t| = |\delta_0| e^{\lambda t}\tag{3}$$

where λ is the Lyapunov exponent (Sprott, 2003). By isolating λ and taking the limit as $t \rightarrow \infty$ and $\delta_0 \rightarrow 0$ to obtain a global indicator of the stability of a dynamical system we get the maximal Lyapunov exponent.

Definition 2.2.1 (Maximal Lyapunov Exponent).

$$\lambda_{\max} = \lim_{t \rightarrow \infty} \lim_{\delta_0 \rightarrow 0} \frac{1}{t} \ln \frac{|\delta_t|}{|\delta_0|}\tag{4}$$

The value of the maximal Lyapunov exponent gives insights on the behaviour of the system. Mainly,

- If $\lambda_{\max} < 0$: it is a dynamical system that converges to a stable fixed point.
- If $\lambda_{\max} = 0$: it is a limit cycle, the dynamical system is stable in sense of Lyapunov.

⁶Usually special cases of Navier-Stokes equations.

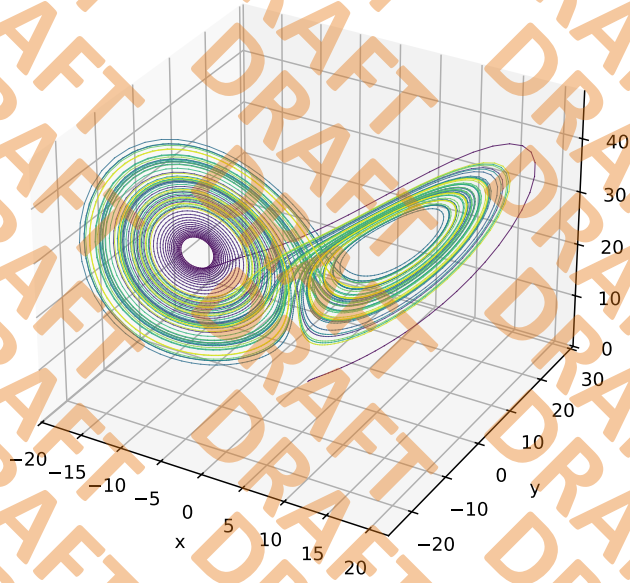


Figure 2 : 3D plot of (2) solved using RK4 scheme. The ordered but chaotic behaviour mentioned at the beginning of this section is clearly seen here.

- If $\lambda_{\max} > 0$: it is an unstable dynamical system with a chaotic behavior.

In discrete dynamical systems $X_{n+1} = f(X_n)$ (or continuous systems that have been discretised for solving) the lyapunov exponent is defined as

$$\lambda_{\max} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \ln |f'(X_n)|. \quad (5)$$

Remark 2.2.2. The formulation (5) comes from the chain rule. Essentially,

$$\delta_N = \tilde{X}_N - X_N = f^N(X_0 + \delta_0) - f^N(X_0) \underset{\delta_0 \rightarrow 0}{\approx} \delta_0 (f^N(X_0))' \underset{\delta_0 \rightarrow 0}{\approx} \delta_0 \prod_{i=0}^{N-1} f'(X_i). \quad (6)$$

So that ($\delta_0 \rightarrow 0$ being implicit as initial perturbations are inherently very small)

$$\lambda_{\max} = \lim_{N \rightarrow \infty} \frac{1}{N} \ln \left| \prod_{i=0}^{N-1} f'(X_i) \right| = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \ln |f'(X_n)|. \quad (7)$$

Definition 2.2.3 (Lyapunov Time). The Lyapunov time is defined as the inverse of the maximal lyapunov exponent. It is the time at which the separation is e times bigger than the initial perturbation.

$$t_e = \frac{1}{\lambda_{\max}} \quad (8)$$

We can also define the error doubling (or tripling etc) time $t_2 = \frac{\ln(2)}{\lambda_{\max}}$ which is the time at which the initial separation has been doubled.

Remark 2.2.4. The maximal Lyapunov exponent can be easily estimated if multiple trajectories starting from closely perturbed initial conditions are available. Given that the separation grows exponentially as $|\delta_t| \approx |\delta_0| e^{\lambda t}$, taking the natural logarithm yields:

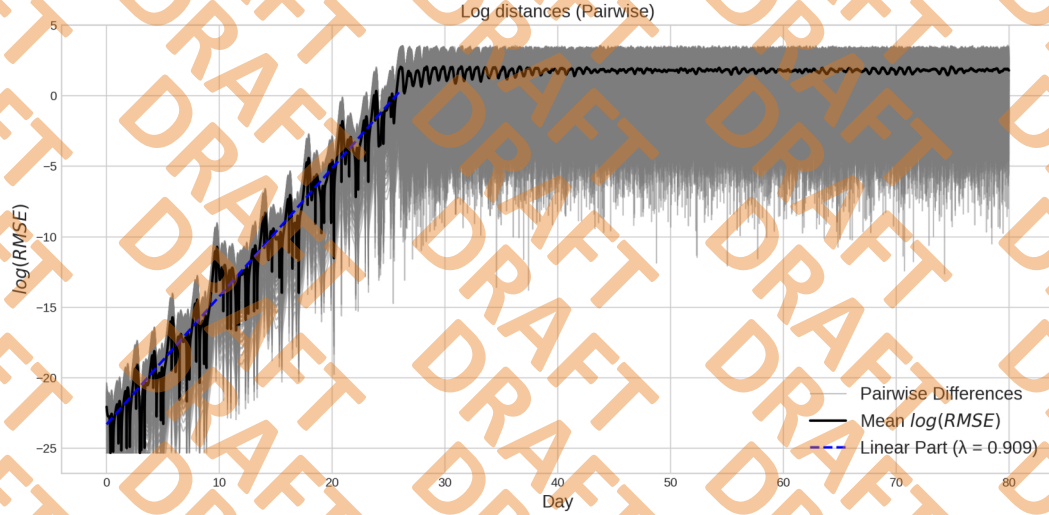


Figure 3 : Plot of all the pairwise log differences between 50 trajectories starting from a random point inside the attractor. The starting point are perturbed as $\tilde{X} = (\tilde{x}, \tilde{y}, \tilde{z}) = X(1 + (10^{-5}, 0, 0) \times \mathcal{U}([- \frac{1}{2}, \frac{1}{2}]))$. The curve exhibits the chaotic behaviour that dampens over time as the system reaches something that is analogous to some stationary climate mean where the trajectories just stay in the attractor but do not have anything in common anymore.

$$\ln|\delta_t| \approx \ln|\delta_0| + \lambda t. \quad (9)$$

By plotting the average of $\ln|\delta_t|$ against time t for several trajectory pairs, $\lambda_{\max}$ can be extracted simply as the slope of the linear growth phase⁷. Note that this linear growth only holds initially. Eventually, the separation saturates as the trajectories become completely uncorrelated. Methods to determine where the linear growth stops are various, we use a variant of the knee-elbow method though one could also find the inflexion point of the time moving average of the mean log-distance.

As an example, we compute in Figure 3 the Lyapunov exponent for the (2) system and get a value close to the 0.9056 often found in literature (Sprott, 2003).

A value for the Lyapunov exponent of the climate system is difficult to find in literature because the value might be dependent on many things such as the model used or the variable, the region observed and even the computation method of the exponent.

As such, even though the method we proposed earlier works well on simple systems such as the (2) system where we recovered the true value for the exponent, (Froude et al., 2013; Lorenz, 1982; Simmons et al., 1994) all use a different method proposed by Lorenz's article for the climate system.

They model the dependence of the root mean square error (RMSE) between a sequence of forecasts on their range t . The growth of error is given by

$$\frac{\partial E}{\partial t} = \alpha E \left(1 - \frac{E}{E_{\infty}} \right) \quad (10)$$

where the parameters of the model are α which is the Lyapunov exponent⁸ and the asymptotic level E_{∞} at which the error saturates.

⁷Since $|\delta_0| < \varepsilon$ and t is big enough for the separation to saturate.

⁸more precisely it is the growth rate of small errors

Remark 2.2.5. This could be written with our notations as $\frac{\partial |\delta_t|}{\partial t} = \lambda |\delta_t| \left(1 - \frac{|\delta_t|}{|\delta_\infty|}\right)$. Notice that without the second term (when the error is small), it reduces exactly to the notation we had for the Lyapunov exponent earlier. Keeping the notation with α is however more relevant to emphasize the context: α is used for the growth rate in a climate systems, while λ refers to the pure mathematical property of the attractor. On the simple Lorenz63 model it overestimates slightly ($\sim 10\%$) the exponent but gives results more plausible on climate systems, while being comparable with the literature. (Bednár et al., 2014) justifies the equivalence between the two quantities.

With this method and some refinements⁹ that are not relevant to the scope of this report. (Bednár et al., 2021) find Lyapunov exponents in the range $[0.32; 0.41] \text{ day}^{-1}$ in the ECMWF forecasting system – which is what we will use as a benchmark for the computation of Lyapunov exponents – by computing them on the 500hPa geopotential height over the Northern Hemisphere. Other authors find error doubling times of around 1.2 to 2.5 days (Froude et al., 2013; Lorenz, 1982; Simmons et al., 1994) which equates to exponents in the same range. Redoing the calculations (examples in Figure S1) using ECMWF IFS Ensemble forecasts we get values in the range of $[0.4; 0.7]$, this is what we will use as reference for AIFS. We also use bootstrap to account for sampling uncertainty and give a 95% confidence interval of the growth rate. All our growth rates will be computed on the region 60°W to 60°E and 15°S to 70°N .

2.3 The AIFS models

This section will aim at introducing the core concepts behind AIFS (Lang et al., 2024) and how it was built and trained. This section will assume the reader has basics of deep learning including CNNs and encoder/processor/decoder architectures. A good introduction is given in (James et al., 2023). It is important to remember that, the different AIFS models do not follow any explicit physical law.

2.3.1 The deterministic model (AIFS-Single)

The Artificial Intelligence Forecasting System (AIFS Single) (Lang et al., 2024) was developed at the European Center For Medium Range Weather Forecasting (ECMWF) in 2024 and launched operationally in the beginning of 2025. The AIFS has a resolution of approximately 0.25° . AIFS is based on a graph neural network (GNN) encoder and decoder, a sliding window transformer processor (Liu et al., 2021), and is trained on ECMWF’s ERA5 re-analysis and fine-tuned with ECMWF’s operational numerical weather prediction (NWP) analyses. AIFS is trained to predict the atmospheric state of the climate at 6 hours timesteps $X(t+6) = \text{AIFS}(X(t), X(t-6))$. It is first trained to predict a single step – by minimising the RMSE between the output and the verifying observation – and then to predict on timescales up to 72 hours by auto-regressively rollout during a second phase.

The use of AIFS in this work is justified by multiple reasons, the first being the computational cost of AI models being orders of magnitude faster than classical NWP. A 30-days forecast using AIFS runs in 2min on a single NVIDIA A100 GPU, compared to hours on a supercomputer for NWP. It is also very skilful and outperforms ECMWF’s Integrated Forecasting System (IFS) with higher anomaly correlations at medium and long range forecast horizons (Lang et al., 2024). Finally, AIFS is completely open source (architecture and weights), which is a crucial requirement for implementing the custom sampling and resampling experiments detailed in this study.

2.3.2 The ensemble model (AIFS-Ens)

While AIFS-Single is highly skilled, it lacks the ability to represent the underlying uncertainty of the predictions it gives. Indeed, in most forecasting tasks, models outputs a single value that ideally

⁹Control variables to correct bias and reduce variance using to a known Lorenz96 model.

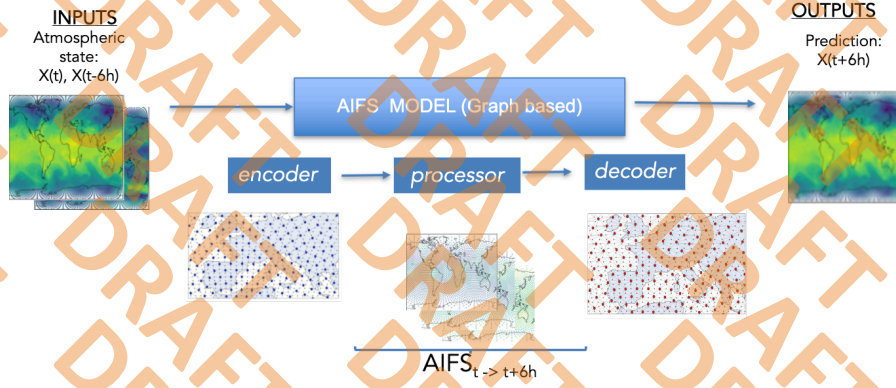


Figure 4 : Schematic view of the AIFS architecture¹⁰. The two atmospheric states at $t - 6, t$ are encoded from a N320 reduced Gaussian grid onto a octahedral O96 grid by a GNN. This latent representation is then processed by a shifted-window transformer (Liu et al., 2021), which applies local self-attention (Vaswani et al., 2023) within bounded regions and shifts these windows across successive layers to efficiently capture spatial dependencies across the entire grid. Finally, the updated latent space on the O96 grid is mapped back to the N320 grid by a second GNN for output.

represents the most likely value of the system at future steps. In this context the prediction can be evaluated using metrics such as the Mean Absolute Error (MAE) or the Root Mean Square Error (RMSE). However, as explained in Section 2.2 the climate system is highly chaotic and therefore a probabilistic forecasting would be more ideal for the task of predicting weather given the uncertainty on initial conditions. This where AIFS-Ens comes in.

The AIFS-Ens (Lang et al., 2026) model is essentially the same model as AIFS-Single in terms of architecture. The difference between the two is in the training procedure and losses used. Unlike the deterministic AIFS-Single, AIFS-Ens training is inherently probabilistic. To generate ensemble spread, independent Gaussian noise is injected via 4 noise channels at each forecast step and model instance (Figure 5). The noise is processed by a two-layer perceptron (MLP) and injected into the model using conditional layer normalizations (Chen et al., 2021) instead of the standard layer normalisations (Ba et al., 2016) in the processor pre-norm transformer layers. This model also uses the afCRPS loss function instead of a regular RMSE, penalising distribution gap rather than mean error (see Sup B for a more detailed explanation of CRPS).

The training of this model is similar to that of the deterministic model, with data up to 2023. Thanks to this procedure for noise injection, inference with AIFS-CRPS is simply running the model instances with different random seeds. Given suitable initial conditions, this can generate as many independent ensemble prediction as needed, without even having to use perturbed ensemble as initial conditions.

2.4 Rare events algorithms

There exists multiple ways to sample from a given distribution. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, we want to estimate $\mathbb{P}(X \in A) = \mathbb{E}[\mathbb{1}_A(X)]$, the probability of a very rare event. Suppose we have a sample $(X_i)_{i=1, \dots, n}$, then the simple Monte Carlo estimator is

$$\hat{\mathbb{P}}(A) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_A(X_i). \quad (11)$$

This estimator is unbiased, consistent and normally asymptotic but converges slowly. We have

¹⁰Graphics from <https://huggingface.co/ecmwf/aifs-single-1.0>

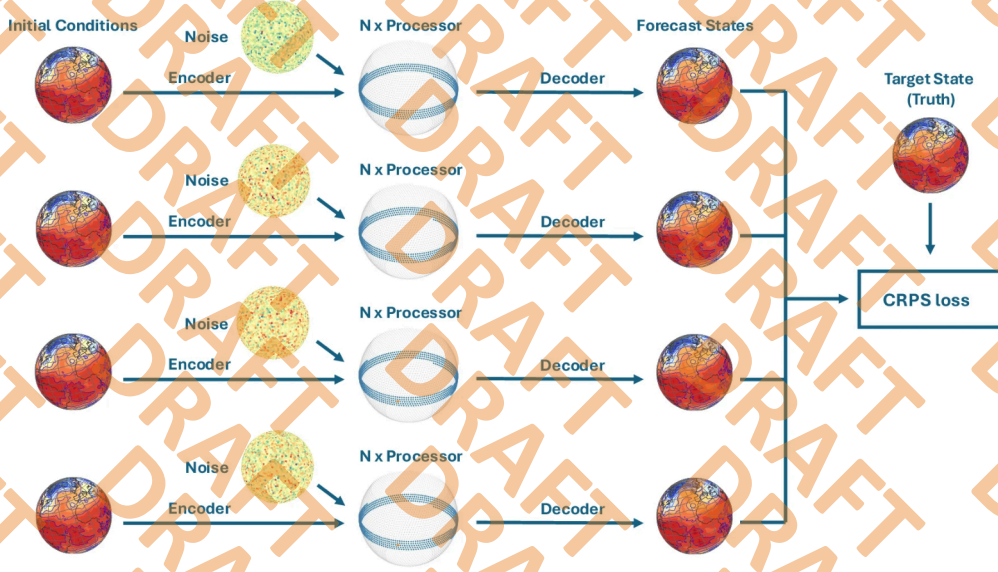


Figure 5 : Training procedure of AIFS-CRPS from (Lang et al., 2026). A small ensemble of atmospheric states is propagated forward in time using separate model instances (that share the same weights). The ensemble forecasts are then gathered and the afCRPS loss is calculated with respect to a single target (ERA5).

$$\mathbb{V}[\hat{\mathbb{P}}(A)] = \frac{\mathbb{V}[\mathbb{1}_A(X)]}{n} = \frac{\mathbb{P}(A)(1 - \mathbb{P}(A))}{n} \underset{\mathbb{P}(A) \text{ small}}{\approx} \frac{\mathbb{P}(A)}{n}. \quad (12)$$

If we were to compute a return level of 1000 years, i.e. $\mathbb{P}(A) = 10^{-3}$ with a relative error of 10% (i.e. a standard deviation of 10^{-4}) we would need $n \approx 10^5$ samples which is a ridiculous amount for most models.

This motivates the use of importance sampling or splitting methods to account for these issues.

2.4.1 Ensemble boosting

Ensemble boosting is an algorithm that allows to generate storylines. Storylines are useful to a posteriori evaluate whether an event could have been more intense. It was proposed in climate sciences by (Gessner et al., 2021) and used in multiple studies (Bloin-Wibe et al., 2025; Fischer et al., 2023; Gessner et al., 2023; Sippel et al., 2024). The idea is to identify extreme events (which are called parent events) in a long simulation or historical data, to then resample these extremes by perturbing initial states from a few days/weeks before they occurred. With a proper lead time before the events, the trajectories (which are called boosted trajectories) will separate because of the chaoticity of the system, this allows sampling events even more extreme than the original. A schematical representation of the algorithm is given in Figure 6.

Originally, this algorithm could not be used to recover the associated probabilities of extremes such as return times, but thanks to the work of (Bloin-Wibe et al., 2025) it is now possible. The idea is inspired from the framework of Subset Simulation (Au & Beck, 2001). One cannot naively compute probabilities of the temperature exceeding the threshold T_{ref} with a crude Monte Carlo estimator on the boosted simulations. Indeed, doing this would give probabilities that are conditioned on the initial conditions at lead time t from the parent events. What we want are the unconditional probabilities, these are recovered by a chain of conditional probabilities.

We note N_{parent} the amount of parent events selected with a threshold of T_{ref} , N_{lead} the number of lead times per parent events and N_{batch} the number of boosted trajectories per parent event and lead

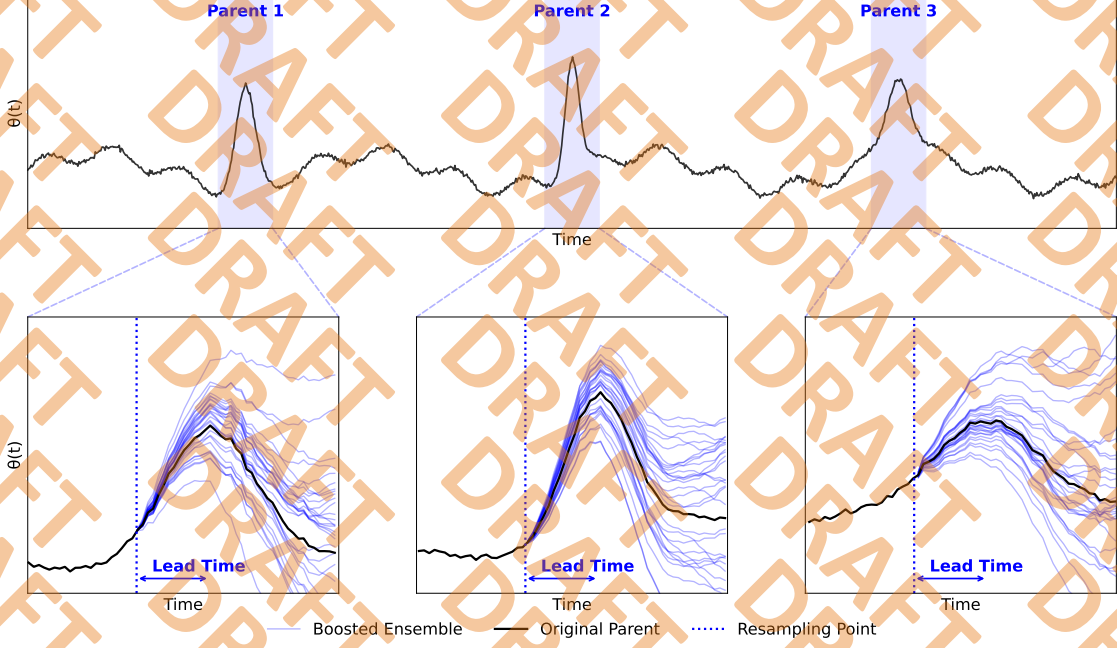


Figure 6 : Schematics of the Ensemble Boosting algorithm. On top, the original time series of an observable θ and 3 highlighted extremes. Below, the 3 extremes are resampled by starting new ensemble simulations a few days/weeks before the occurrence of the extremes.

time. This gives a total amount of boosted trajectories $N_b = N_{\text{lead}} \cdot N_{\text{batch}} \cdot N_{\text{parent}}$. We also note $T = \{T^n | n = 1, 2, \dots, N\}$ the temperatures of the parent events, $T_b = \{T_b^m | m = 1, 2, \dots, N_b\}$ the temperatures of the boosted trajectories. The parent events antecedent conditions are $AC_t^0 = \{X_t | T_{t=0} \geq T_{\text{ref}}\}$ and the boosted trajectories integrated for a duration f (forward time) are $AC_t^\omega = \{Y | Y = \mathcal{M}_f(X_t), X_t \in AC_t^0\}$. This last notation highlights that the variability can come from either initial perturbations or the model $\mathcal{M}$ used itself.

Using conditional probabilities, we have, with $T_{\text{ext}} \geq T_{\text{ref}}$

$$\begin{aligned} \mathbb{P}(T_{t=0} \geq T_{\text{ref}} \cap AC_t^\omega) &= \mathbb{P}(T_{t=0} \geq T_{\text{ref}} | AC_t^\omega) \mathbb{P}(AC_t^\omega), \\ \mathbb{P}(T_{t=0} \geq T_{\text{ext}} \cap AC_t^\omega) &= \mathbb{P}(T_{t=0} \geq T_{\text{ext}} | AC_t^\omega) \mathbb{P}(AC_t^\omega). \end{aligned} \quad (13)$$

Since $T_{\text{ext}} \geq T_{\text{ref}}$ and $AC_t^0 \subset AC_t^\omega$ the LHS probabilities simplify to $\mathbb{P}(T_{t=0} \geq T_{\text{ref}})$ and $\mathbb{P}(T_{t=0} \geq T_{\text{ext}})$. The subscript $t = 0$ indicates that it is the probability at the day of the parent event's peak. We assume this probability is the same for each event as we assume no seasonality and stationary data¹¹. Therefore, these probabilities are the climatological probabilities, i.e. $\mathbb{P}(T_{t=0} \geq T_{\text{ref}}) = \mathbb{P}(T \geq T_{\text{ref}})$ and $\mathbb{P}(T_{t=0} \geq T_{\text{ext}}) = \mathbb{P}(T \geq T_{\text{ext}})$.

The probability of reaching a temperature greater than T_{ext} is

$$\mathbb{P}(T \geq T_{\text{ext}}) = \mathbb{P}(T \geq T_{\text{ref}}) \frac{\mathbb{P}(T \geq T_{\text{ext}} | AC_t^\omega)}{\mathbb{P}(T \geq T_{\text{ref}} | AC_t^\omega)} \quad (14)$$

and can be estimated as

$$\hat{\mathbb{P}}_{T \geq T_{\text{ext}}} = \left(\frac{1}{N} \sum_{n=1}^N 1_{\{T^n \geq T_{\text{ref}}\}} \right) \frac{\frac{1}{N_b} \sum_{m=1}^{N_b} 1_{\{T_b^m \geq T_{\text{ext}}\}}}{\frac{1}{N_b} \sum_{m=1}^{N_b} 1_{\{T_b^m \geq T_{\text{ref}}\}}}. \quad (15)$$

¹¹These are strong assumptions but using detrended summer maxima for parent events satisfies them.

The first term represents the proportion of event reaching the threshold in the baseline data, the numerator and denominator of the fraction are respectively the proportion of boosted trajectories that reach T_{ext} or T_{ref} .

In the original paper (Bloin-Wibe et al., 2025), the authors also show that under some mild assumptions the estimator is unbiased and that it reduces errors compared to a naive Monte Carlo estimator.

2.4.2 Iterative boosting

Taking advantage of the speed improvements of AI weather models over physics models, we developed a new sampling method, Iterative Ensemble Boosting. This method (Figure 7) allows to push the model further into high temperatures. It works by iterating the boosting multiple times with different lead times and forward times as we get further in the tail of the distribution (so that the trajectory is the same size in the end as if we did simple boosting, for example 30 days max with AIFS-Ens), selecting the top $q\%$ trajectories that maximized the observable in the previous phase. The final trajectories can be reconstructed by merging the trajectories with their parent for each phase and overriding the older phases with the latest ones when they overlap.

Using notations (Table 2) similar to and formulas from Section 2.4.1 we start by we can estimating easily the climate probability $\mathbb{P}(T \geq T_{\text{ref},0})$

$$\hat{\mathbb{P}}_{T \geq T_{\text{ref},0}} = \frac{1}{N} \sum_{n=1}^N 1_{\{T^n \geq T_{\text{ref},0}\}}. \quad (19)$$

Then, the first phase of iterative boosting consists in a first boosting. Integrating the model $\mathcal{M}$ forward for a duration f_1 from the antecedent states found at time t_1 before the event gives $\text{AC}_{t_1}^\omega = \{Y \mid Y = \mathcal{M}_{f_1}(X_{t_1}), X_{t_1} \in \text{AC}_{t_1}^0\}$. Therefore, we have, as for the simple boosting

$$\mathbb{P}(T \geq T_{\text{ref},1}) = \mathbb{P}(T \geq T_{\text{ref},0}) \frac{\mathbb{P}(T \geq T_{\text{ref},1} \mid \text{AC}_{t_1}^\omega)}{\mathbb{P}(T \geq T_{\text{ref},0} \mid \text{AC}_{t_1}^\omega)} \quad (20)$$

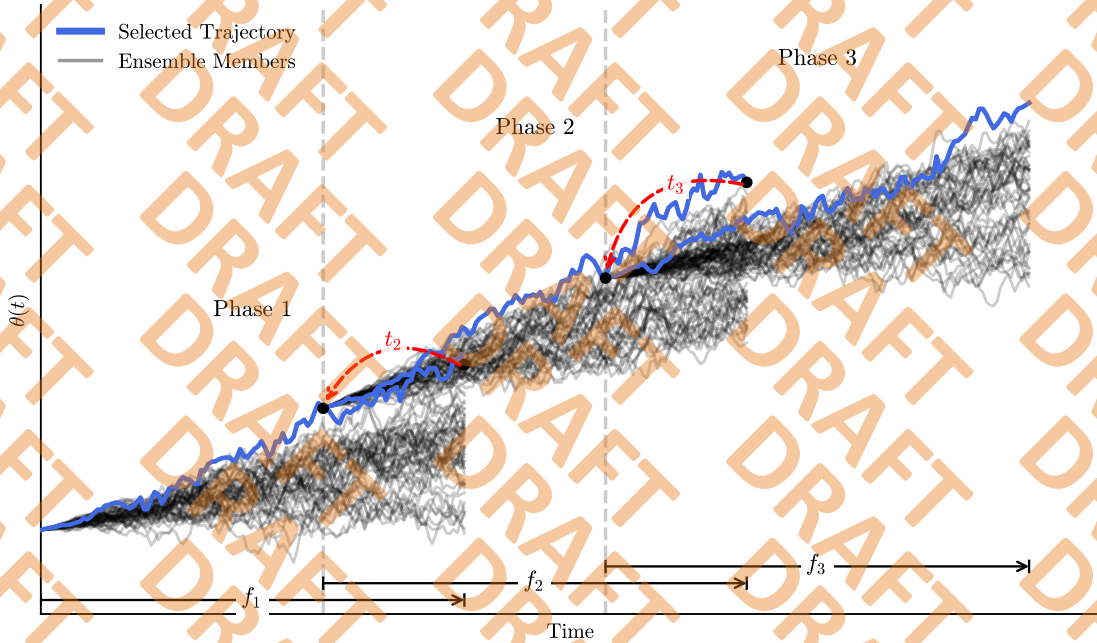


Figure 7 : Schematics of the iterative boosting algorithm with q such that only one trajectory is selected for visualisation purposes.

Table 2 : Definitions and descriptions of important quantities used for ensemble boosting.

$\{t_j\}_{j=1,\dots,J}$	the different lead times
$\{f_j\}_{j=1,\dots,J}$	the different forward integration times (duration of the model run)
N	same as in normal boosting (number of years in historical)
$\{N_{b,j}\}_{j=1,\dots,J}$	the different amounts of boosted members
$\{T_{\text{ref},j}\}_{j=1,\dots,J}$	the different thresholds, ($T_{\text{ref},0}$ is the user defined first threshold and $T_{\text{ref},J}$ will be T_{ext})
$AC_{t_j}^0$	set of antecedent condition of boosted members of phase j from which T_{b_j} of phase j are sampled. $AC_{t_1}^0 = \{X_{t_1} \mid T_{t=0} \geq T_{\text{ref},0}\}, \quad j = 1 \quad (16)$ (states obtained by going back t_1 from the historical events max) $AC_{t_j}^0 = \{X_{t_j} \mid T_b^{m,j-1} \geq T_{\text{ref},j-1}\}, \quad j > 1 \quad (17)$ (states obtained by going back t_j from the end of the successful trajectories of phase $j-1$)
$AC_{t_j}^\omega$	set of boosted trajectories of phase j generated by integrating forward for a duration f_j, ω represents the perturbation either on initial conditions or in the model itself $AC_{t_j}^\omega = \{Y \mid Y = \mathcal{M}_{f_j}(X_{t_j}), X_{t_j} \in AC_{t_j}^0\} \quad (18)$
$\mathcal{M}$	A model, for example AIFS-Ens
$N_{\text{batch},j}$	the number of model runs (branches) run from each selected trajectory during phase j
q	fraction of top trajectories kept at the end of each phase
$T_b^{m,j}$	maxima for the observable on the m -th boosted member of phase j

which can be estimated by

$$\Rightarrow \hat{\mathbb{P}}_{T \geq T_{\text{ref},1}} = \hat{\mathbb{P}}_{T \geq T_{\text{ref},0}} \frac{\frac{1}{N_{b,1}} \sum_{m=1}^{N_{b,1}} 1_{\{T_b^{m,1} \geq T_{\text{ref},1}\}}}{\frac{1}{N_{b,1}} \sum_{m=1}^{N_{b,1}} 1_{\{T_b^{m,1} \geq T_{\text{ref},0}\}}} \quad (21)$$

We select the top $q\%$ for the next phase so that $\frac{1}{N_{b,1}} \sum_{m=1}^{N_{b,1}} 1_{\{T_b^{m,1} \geq T_{\text{ref},1}\}} = q$. This gives

$$\hat{\mathbb{P}}_{T \geq T_{\text{ref},1}} = \hat{\mathbb{P}}_{T \geq T_{\text{ref},0}} \frac{q}{\frac{1}{N_{b,1}} \sum_{m=1}^{N_{b,1}} 1_{\{T_b^{m,1} \geq T_{\text{ref},0}\}}} \quad (22)$$

Doing so gives $AC_{t_2}^0 = \{X_{t_2} \mid T_b^{m,1} \geq T_{\text{ref},1}\}$ after going back at time t_2 before the end of the successful trajectories of phase 1. We then reiterate, integrating forward $\mathcal{M}$ for a duration f_2 from these new antecedent states and obtaining $AC_{t_2}^\omega$. The probabilities can be chained and give

$$\begin{aligned}
\mathbb{P}(T \geq T_{\text{ref},2}) &= \mathbb{P}(T \geq T_{\text{ref},1}) \frac{\mathbb{P}(T \geq T_{\text{ref},2} \mid \text{AC}_{t_2}^\omega)}{\mathbb{P}(T \geq T_{\text{ref},1} \mid \text{AC}_{t_2}^\omega)} \\
&= \mathbb{P}(T \geq T_{\text{ref},0}) \frac{\mathbb{P}(T \geq T_{\text{ref},1} \mid \text{AC}_{t_1}^\omega)}{\mathbb{P}(T \geq T_{\text{ref},0} \mid \text{AC}_{t_1}^\omega)} \frac{\mathbb{P}(T \geq T_{\text{ref},2} \mid \text{AC}_{t_2}^\omega)}{\mathbb{P}(T \geq T_{\text{ref},1} \mid \text{AC}_{t_2}^\omega)}.
\end{aligned} \tag{23}$$

We select the top $q\%$ again and this can be estimated as

$$\hat{\mathbb{P}}_{T \geq T_{\text{ref},2}} = \hat{\mathbb{P}}_{T \geq T_{\text{ref},0}} \frac{q}{\frac{1}{N_{b,1}} \sum_{m=1}^{N_{b,1}} 1_{\{T_b^{m,1} \geq T_{\text{ref},0}\}}} \frac{q}{\frac{1}{N_{b,2}} \sum_{m=1}^{N_{b,2}} 1_{\{T_b^{m,2} \geq T_{\text{ref},1}\}}} \tag{24}$$

$$= \hat{\mathbb{P}}_{T \geq T_{\text{ref},0}} \frac{q^2}{\prod_{j=1}^2 \frac{1}{N_{b,j}} \sum_{m=1}^{N_{b,j}} 1_{\{T_b^{m,j} \geq T_{\text{ref},j-1}\}}}. \tag{25}$$

This methodology is reiterated J times yielding

$$\mathbb{P}(T \geq T_{\text{ref},J}) = \mathbb{P}(T \geq T_{\text{ref},0}) \frac{\prod_{j=1}^J \mathbb{P}(T \geq T_{\text{ref},j} \mid \text{AC}_{t_j}^\omega)}{\prod_{j=1}^J \mathbb{P}(T \geq T_{\text{ref},j-1} \mid \text{AC}_{t_j}^\omega)} \tag{26}$$

which is estimated as

$$\hat{\mathbb{P}}_{T \geq T_{\text{ref},J}} = \left(\frac{1}{N} \sum_{n=1}^N 1_{\{T^n \geq T_{\text{ref},0}\}} \right) \frac{q^J}{\prod_{j=1}^J \frac{1}{N_{b,j}} \sum_{m=1}^{N_{b,j}} 1_{\{T_b^{m,j} \geq T_{\text{ref},j-1}\}}}. \tag{27}$$

In order to better approximate the very tail of the distribution we take $T_{\text{ext}} \geq T_{\text{ref},J-1}$ and estimate $\mathbb{P}(T > T_{\text{ext}})$ at phase J ¹²

$$\hat{\mathbb{P}}_{T \geq T_{\text{ext}}} = \left(\frac{1}{N} \sum_{n=1}^N 1_{\{T^n \geq T_{\text{ref},0}\}} \right) \left(\frac{q^{J-1}}{\prod_{j=1}^{J-1} \frac{1}{N_{b,j}} \sum_{m=1}^{N_{b,j}} 1_{\{T_b^{m,j} \geq T_{\text{ref},j-1}\}}} \right) \left(\frac{1}{N_{b,J}} \sum_{m=1}^{N_{b,J}} 1_{\{T_b^{m,J} \geq T_{\text{ext}}\}} \right). \tag{28}$$

In theory one could choose any J but in practice, when using AIFS-Ens we use $J < 5$ for stability reasons and because we need the model to diverge. A pseudo-code for the algorithm is given in Algorithm 1.

An important implementation detail lies in the idea that we should keep $N_{b,j} = N_b$ constant over the phases (we cannot just run $N_{\text{batch}} = 100$ for each trajectory selected because the number of trajectories would grow exponential), we can write that the total number of boosted simulations of phase j is the the amount of selected trajectories from the preceding phase multiplied by how much runs we make with each, that is $N_{b,j} = N_{b,j-1} \times q \times N_{\text{batch},j}$ since we want to keep the total amount constant that means $q \times N_{\text{batch},j} = 1$. This is the tradeoff we have to play with. If $q \searrow$ and $N_{\text{batch},j} \nearrow$, the events that are resampled are more rare, and are resampled more times. But a q too small might break the computation

¹²This can also be used at any phase.

of probability, or imply a not diverse enough representation of the antecedent conditions (but from the typicality argument of (Noyelle et al., 2023a) this is not an issue). This configuration could also, if parallelization is implemented per batch, to use less GPUs. On the other hand, if $q \nearrow$ and $N_{\text{batch},j} \searrow$, the events that are resampled are less rare, and are resampled less times. The set of antecedent conditions is more diverse and it might be more gentle on the chained probabilities. That being said, it might also not end up as far in the tail. This configuration would be more expensive to parallelize.

To validate the performances of our algorithm, we test it on a toy Ornstein-Uhlenbeck process. Our toy system is defined for $X_t \in \mathbb{R}$ by the SDE $dX_t = -X_t dt + dW_t$ and integrated via a Runge-Kutta method. The drift term $-X_t dt$ is negative, pulling the system back to 0 and the diffusion term dW_t is a standard brownian motion. We first run a control set of trajectories ($N = 1,500,000$) to estimate a ground truth for the maximum of the OU process¹³. Then we perform a naive Monte Carlo estimation on a small set of trajectories ($N = 200$), from that small set of trajectories we perform ensemble and iterative boosting as well. We make sure the computation cost (number of integration steps) is the same for all methods. The ensemble boosting and iterative boosting are performed with $N_{\text{parent}} = 20$, then the respective parameters are $N_b = 560$ for the boosting and $N_b = 400$ (with $q = 0.05$) for the iterative boosting. We use a lead time of $t = 0.2$ for the boosting while we use $f_i = [0.4 \ 0.4 \ 0.6]$ and $t_i = [0.2 \ 0.2 \ 0.2]$ for iterative boosting. This way all the methods are comparable in terms of compute needs, we do not account for the construction of the small trajectory in the cost for the two boosting methods as in our real use case this data will already be available, we need its cost for comparison with a purely naive method though. We also fit a GEV (Coles, 2001) on the smaller set for comparison. GEVs are statistical models used to model block maxima (annual here) of random variables.

We find that iterative boosting performs quite well (Figure 8), being on par with the control run further in the tail than a naive Monte Carlo. It gives a better estimate of the return times compared to the GEV, which severely underestimates the return values. The GEV has a negative shape parameter, implying

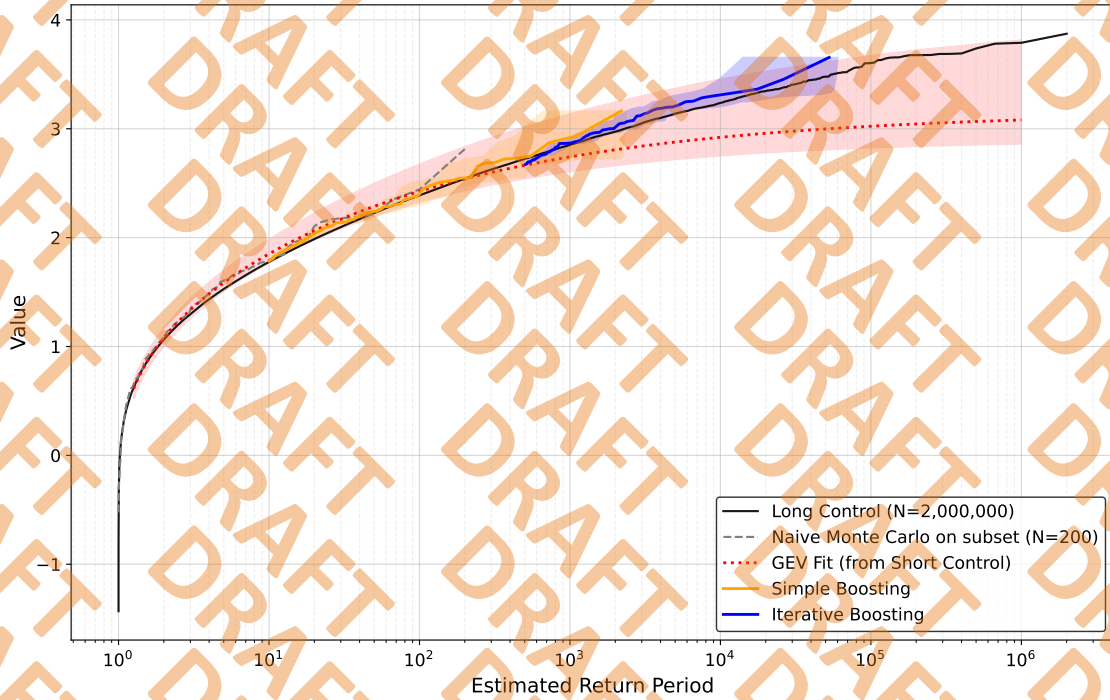


Figure 8 : Returns times estimated for the maximum of an OU process. Comparison of a Naive Estimation with a GEV, the boosting estimator and the iterative boosting estimator.

¹³It amounts to doing it on a very long trajectory, and taking block maxima but this approach allows for parallelising

that return values are bounded, while the maxima of the OU are not. The advantage of both boosting methods in the context of climate science does not lie only in a better estimation of return times, but also in the ability to retrieve the trajectories of the system studied. If we quantify the efficiency of the algorithms solely by calculating the total number of integration steps required per unit of maximum return time: $\frac{N_{\text{compute}}}{\max(R)}$. We find efficiency improvements of the simple boosting over the naive monte carlo of up to 15 times, and of up to 40 times for iterative boosting against ensemble boosting, meaning an improvement of iterative boosting over the naive estimation of up to 600 times.

Require: Historical data, total phases J (with t_j and f_j), selection quantile q , first threshold $T_{\text{ref},0}$

1 Compute the baseline probability:

$$\hat{\mathbb{P}}_{T \geq T_{\text{ref},0}} \leftarrow \frac{1}{N} \sum_{n=1}^N 1_{\{T^n \geq T_{\text{ref},0}\}} \quad (29)$$

2 Select initial states by going back t_1 from the historical events exceeding the first threshold:

$$AC_{t_1}^0 \leftarrow \{X_{t_1} \mid T_{t=0} \geq T_{\text{ref},0}\} \quad (30)$$

3 **for** $j = 1$ **to** $J - 1$ **do**

4 Run the model forward f_j from the antecedent states to generate the boosted ensemble:

$$AC_{t_j}^\omega \leftarrow \{Y \mid Y = \mathcal{M}_{f_j}(X_{t_j}), X_{t_j} \in AC_{t_j}^0\} \quad (31)$$

5 Evaluate the observable (max) $T_b^{m,j}$ for all N_b ensemble members

6 Find the new threshold $T_{\text{ref},j}$ such that exactly the top $q\%$ of trajectories exceed it

7 Update the chained probability estimate:

$$\hat{\mathbb{P}}_{T \geq T_{\text{ref},j}} \leftarrow \hat{\mathbb{P}}_{T \geq T_{\text{ref},j-1}} \frac{q}{\frac{1}{N_b} \sum_{m=1}^{N_b} 1_{\{T_b^{m,j} \geq T_{\text{ref},j-1}\}}} \quad (32)$$

8 Go back t_{j+1} from the end of the successful trajectories to build the antecedent conditions for the next phase:

$$AC_{t_{j+1}}^0 \leftarrow \{X_{t_{j+1}} \mid T_b^{m,j} \geq T_{\text{ref},j}\} \quad (33)$$

9 **end**

10 Run the model forward f_J from the final antecedent states: $AC_{t_J}^\omega \leftarrow \{Y \mid Y = \mathcal{M}_{f_J}(X_{t_J}), X_{t_J} \in AC_{t_J}^0\}$

11 Evaluate the observable (max) $T_b^{m,J}$ against the extreme target T_{ext}

12 Compute the final probability estimate of the extreme event:

$$\hat{\mathbb{P}}_{T \geq T_{\text{ext}}} \leftarrow \left(\frac{1}{N} \sum_{n=1}^N 1_{\{T^n \geq T_{\text{ref},0}\}} \right) \left(\frac{q^{J-1}}{\prod_{j=1}^{J-1} \frac{1}{N_{b,j}} \sum_{m=1}^{N_{b,j}} 1_{\{T_b^{m,j} \geq T_{\text{ref},j-1}\}}} \right) \left(\frac{1}{N_{b,J}} \sum_{m=1}^{N_{b,J}} 1_{\{T_b^{m,J} \geq T_{\text{ext}}\}} \right) \quad (34)$$

13 Reconstruct the full trajectories by merging phases 1 to J , overriding older data with the latest phases when they overlap

14 **return** $\hat{\mathbb{P}}_{T \geq T_{\text{ext}}}$ and reconstructed trajectories

Algorithm 1 : Iterative Ensemble Boosting.

Finally, this algorithm, even though inspired by a variety of existing algorithms differs (to the best of my knowledge) from what already exists in literature. The algorithm improves the 1-step approach proposed by ensemble boosting (Bloin-Wibe et al., 2025; Fischer et al., 2023; Gessner et al., 2021) by reaching further into the tail of the distribution without increasing compute needs. Adaptive multilevel splitting (AMS) (Cérou et al., 2019; Cérou & Guyader, 2007; Ragone et al., 2018) and its trajectory based variant TAMS (Lestang et al., 2018) split the trajectories at the exact time a threshold is crossed, which is not viable for transient extremes such as temperature peaks. Cloned trajectories need some lead time to diverge. Our algorithm resamples from saved antecedent states (early splitting), fixing this issue. Another alternative is Trying-Early AMS (TEAMS) (Finkel & O'Gorman, 2024), this approach uses an accept/reject step from (Au & Beck, 2001). In this approach of boosting one parent with a batch of 1 repeatedly until reaching the desired extremes, the number of rejections is unknown. The sequential nature of this algorithm limits parallelization and computational resources are inherently wasted on rejected trajectories. Our algorithm combines the early splitting/perturbation from TEAMS and ensemble boosting and the adaptive threshold and selection from AMS and TAMS. This allows reaching deeper into the tail than 1-step ensemble boosting, without the computational hassle of trial-and-error rejection loops. While rejected trajectories in TEAMS do contribute to the probability estimate, they do not generate new physical states. In contrast, our approach utilizes every trajectory to compute the threshold and offers a diverse set of extremes, while having a fixed cost.

2.5 Physics of heatwaves

With rare events algorithms, one can generate extreme temperatures. But what are the underlying processes being heatwaves, and how far are we from the physical limits of maximum temperature? This parts aims at introducing key physical principles with respect to heatwaves.

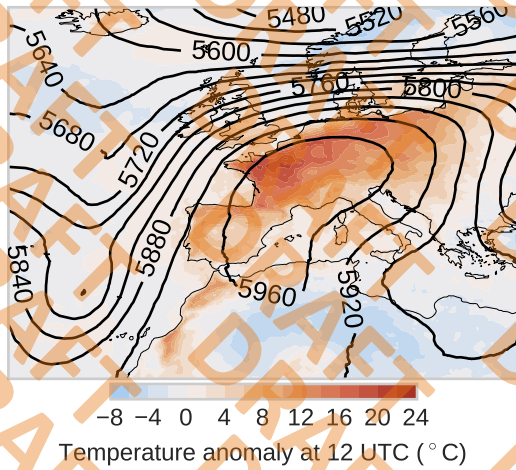
2.5.1 Synoptic processes of heatwaves.

It has been showed (Noyelle et al., 2023a) that the variance between trajectories leading to extreme events diminishes with the level of extreme, in the IPSL-CM6A-LR model (Boucher et al., 2020). In other terms, the more extreme an event is, the more likely it is that the atmospheric conditions associated to it are similar.

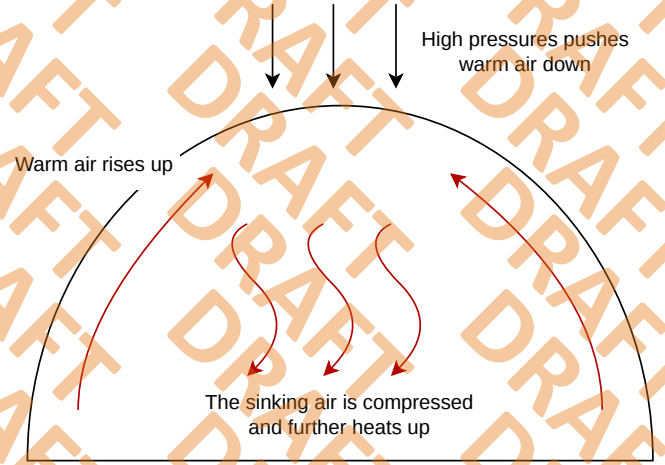
European summer heatwaves are events primarily caused by known drivers. These drivers are characterised by the combination of very large synoptic systems known as Omega blockings Figure 9a (due to their particular shape that resembles the Greek letter Ω) and low soil moisture. The omega blockings create so-called heat domes Figure 9b. Warm air from the south is brought north. In the center of the high pressure system, warm air at the surface rises up in the atmosphere, but high atmospheric pressure forces it to sink (subsidence), compressing it and further heating the surface. The blocking also stops clouds outside of the dome, preventing them from protecting the surface from solar radiation. Lastly, this high-pressure system also diverts the jet stream over the blocking and stops it from displacing high altitude warm air to the east.

2.5.2 Convective instability

In order to estimate the maximal temperature a heatwave can reach, (Zhang & Boos, 2023) hypothesised and showed empirically that the maximal reachable temperature is limited by convective instability that stops the further development of heatwaves. Precisely, over mid latitude lands, the surface air temperature can only rise until precipitation is triggered by the convective instability. This explain why the study of vertical columns is important for heatwaves.



(a) Example of an omega block.



(b) The heat dome.

Figure 9 : Schematics of the processes behind heatwaves.

Convective instability happens when the moist static energy at the surface (MSE_s) reaches higher values than the saturation moist static energy at 500hPa (MSE_{500}^*). That is $MSE_s \geq MSE_{500}^*$.

Moist static energy is used as a measure of the energy conserved for hydrostatic, moist adiabatic vertical motions and useful for understanding moist convection. It is obtained as

$$\begin{aligned} MSE_s &= c_p T_s + L_v q_s + g z_s \\ MSE_{500}^* &= c_p T_{500} + L_v q_{sat}(T_{500}) + g z_{500} \end{aligned} \quad (35)$$

where c_p is the specific heat of air at constant pressure, L_v is the latent heat of vaporisation and g is the gravitational acceleration. The saturation specific humidity at T_{500} is approximated by

$$q_{sat}(T_{500}) \approx \frac{\epsilon e_{sat}(T_{500})}{500 \text{ hPa}} \quad (36)$$

where ϵ is the ratio between water vapour and dry air and $e_{sat}(T_{500})$ is the saturation vapour pressure given by August–Roche–Magnus formula¹⁴

$$e_{sat}(T) = 6.1094 \exp\left(\frac{17.625 \times T}{243.04 + T}\right) \quad (37)$$

where T is in °C and the saturation vapour pressure in hPa.

2.5.3 Decomposition of heat change from a eulerian perspective.

To understand the magnitude of a heat wave we want to derive an equation decomposing temperature change into thermodynamics components. This derivation is partly based on and inspired by from (Holton & Hakim, 2013; Hotz et al., 2024).

Let us remind the ideal gas law:

$$pV = nRT \Leftrightarrow p\alpha = RT \quad (38)$$

where p is the atmospheric pressure (Pa), T is the temperature (K), α is the specific volume (m^3/kg), and $R = 287.05 \text{ J}/(\text{kg}\cdot\text{K})$ is the specific gas constant for dry air.

¹⁴This equation is derived from the Clausius–Clapeyron equation for water vapor under typical atmospheric conditions.

The internal energy per mass unit of dry air is:

$$dU = c_v dT \quad (39)$$

where $c_v = 718 \text{ J/(kg}\cdot\text{K)}$ is the specific heat at constant volume.

And the work done by the system on a mass unit of dry air is:

$$\delta W = p d\alpha. \quad (40)$$

We know from the first principle of thermodynamics that

$$\delta Q = dU + \delta W. \quad (41)$$

Substituting our definitions for internal energy and work:

$$\begin{aligned} \delta Q &= c_v dT + p d\alpha \\ \Rightarrow \frac{Dq}{Dt} &= c_v \frac{DT}{Dt} + p \frac{D\alpha}{Dt}. \end{aligned} \quad (42)$$

By differentiating the ideal gas law (38), we obtain a relation for the work term:

$$p \frac{D\alpha}{Dt} + \alpha \frac{Dp}{Dt} = R \frac{DT}{Dt}. \quad (43)$$

Substituting this back into our heat equation, and noting $c_p = R + c_v$ the specific heat at constant pressure ($1004 \text{ J/(kg}\cdot\text{K)}$), we get:

$$\begin{aligned} \frac{DQ}{Dt} &= (R + c_v) \frac{DT}{Dt} - \alpha \frac{Dp}{Dt} \\ &= c_p \frac{DT}{Dt} - \alpha \frac{Dp}{Dt}. \end{aligned} \quad (44)$$

This formulation is Lagrangian in the sense that to use it we have to be able to follow a parcel of air. This requires the calculation of backward trajectories, which is not to the scope of this report. We want to switch to a Eulerian perspective. In meteorology, the vertical variable is rarely z but rather the pressure p . Expanding the total derivative using the chain rule yields:

$$\begin{aligned} \frac{DT}{Dt} &= \frac{\partial T}{\partial t} + \frac{Dx}{Dt} \frac{\partial T}{\partial x} + \frac{Dy}{Dt} \frac{\partial T}{\partial y} + \frac{Dp}{Dt} \frac{\partial T}{\partial p} \\ &= \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + \omega \frac{\partial T}{\partial p} \\ &= \frac{\partial T}{\partial t} + V_h \cdot \nabla_h T + \omega \frac{\partial T}{\partial p}. \end{aligned} \quad (45)$$

Where u and v are the zonal and meridional wind (m/s), we noted $\omega = \frac{Dp}{Dt}$ the vertical component of wind speed in pressure coordinates¹⁵ (Pa/s).

Substituting this expansion into (44) (and replacing $\alpha = \frac{RT}{p}$):

$$\frac{DQ}{Dt} = c_p \left(\frac{\partial T}{\partial t} + V_h \cdot \nabla_h T \right) + c_p \omega \frac{\partial T}{\partial p} - \frac{RT}{p} \omega. \quad (46)$$

The left side of the equation represents a heating rate in $\text{J/(kg}\cdot\text{s)}$. We would like to have it in (K/s) which is more intuitive, for that we divide the entire equation by c_p . Noting the diabatic heating rate

¹⁵Because pressure increases closer to the surface, a parcel sinking downwards into higher pressure has $\omega > 0$.

as $Q_{\text{diab}} = \frac{DQ}{Dt} \frac{1}{c_p}$ and the Poisson constant as $\kappa = \frac{R}{c_p} \approx 0.286$, we get the following decomposition by rearranging to isolate $\frac{\partial T}{\partial t}$:

$$\frac{\partial T}{\partial t} = Q_{\text{diab}} - V_h \cdot \nabla_h T + \omega \left(\frac{\kappa T}{p} - \frac{\partial T}{\partial p} \right). \quad (47)$$

(47) gives the instantaneous rate of change. If we want to study the total energy balance of a specific meteorological event, we can integrate over time at a specific gridpoint from the start t_s to the end t_e of said event:

pas sur de mettre la version intégrée, je ne m'en sert pas au final, autant l'enlever et décrire directement (47)?, en plus ça me remettrait tout pile à 30 pages

$$\begin{aligned} \int_{t_s}^{t_e} \frac{\partial T}{\partial t} dt &= \underbrace{\int_{t_s}^{t_e} Q_{\text{diab}} dt}_{\text{DIABATIC}} \\ &+ \underbrace{\int_{t_s}^{t_e} [-V_h \cdot \nabla_h T] dt}_{\text{ADVECTION}} \\ &+ \underbrace{\int_{t_s}^{t_e} \omega \left[\begin{array}{cc} \text{ADIABATIC COMPRESSION} & \text{VERTICAL DISPLACEMENT} \\ \frac{\kappa T}{p} & - \frac{\partial T}{\partial p} \end{array} \right] dt}_{\text{VERTICAL}} \end{aligned} \quad (48)$$

Each of these terms represents a distinct physical process: the diabatic term corresponds to heat exchanges such as the absorption of infrared radiation or release of latent heat. The advection term corresponds to heat physically brought in by the horizontal wind. The adiabatic compression term corresponds to heat generated by the air getting compressed under high pressures. The vertical displacement is a term fighting against adiabatic compression. It represents the air being vertically displaced by the wind.¹⁶ Note that in AIFS, we do not explicitly compute the diabatic term as it is often not simple to estimate and decide to not take it into account. Because we know the rate of temperature change we could estimate it by isolating it but this could be a poor estimation.

3 RESULTS

This section aims at presenting the most important results from my internship, concerning the May 2026 heatwave and a test case of the iterative boosting for summer in June, July, August (JJA) temperatures. Results concerning the study of the statistical and physical behaviour of AIFS-Single and AIFS-Ens are available in Sup D. We can summarize the supplementary materials results here by noting that AIFS-Single was not sufficient in terms of chaoticity for our use cases but that AIFS-Ens, was perfectly capable both in terms of physical and statistical behaviour.

3.1 The May 2026 heatwave

In May 2026, France and the UK experienced a record-breaking hot event as daily maximum temperature anomalies¹⁷ of more than 15 °C took local authorities by surprise. While in public discourse, near-surface heat was attributed to an unprecedented high-pressure system over the heatwave region, we suspect that

¹⁶Air in the higher levels is colder.

¹⁷Difference with the season or monthly average.

the concept of a heat dome falls short in explaining the full amplitude of the event. Via perturbed large ensemble medium range forecasts, we investigate the role of a cut-off low over the Iberian Peninsula to explain the temperatures. Elements of this section will be used for an article about this heatwave in preparation with Maeve Mayer, Robin Noyelle, Belinda Hotz, Soulivanh Thao and Pascal Yiou.

We consider an ensemble of 201 AIFS-Ens (Lang et al., 2026) simulations that are initiated on May 17th and 18th. We analyze the simulations for which the z_{500} reaches the 5683m threshold that yields the top 10 near-surface temperature values at 12UTC over the target area (ENS1), and the top 10 coldest (ENS2).

Of all the simulations starting on the 17th and 18th, respectively 17% and 22% exceed a 31.73°C threshold, which is ERA5 May maximum at 12UTC for the Ile de France grid point. Respectively 60% and 82% of the simulations reach the threshold 5683m over the target region.

When comparing (ENS1) and (ENS2) in Figure 10 we see that all members show a high pressure system/omega block over the whole Western Europe. This blocking seems insufficient to be the only reason of high temperature given the difference between the two sets of up to 12°C Figure 10 c. The (ENS1) members all show a cut-off low west of the Iberian Peninsula, anchoring the Omega block and further amplifying its shape and making it more distinct, when the (ENS2) members do not. We hypothesize that the cut-off low in ENS1 generates a counter-clockwise flow and the block's high clockwise flow help transport more Saharian and Iberian hot air into France and the UK. ENS2 lacks the cut-off low; thus, the atmospheric flow transports more ocean air from the Atlantic, which is inherently colder than North

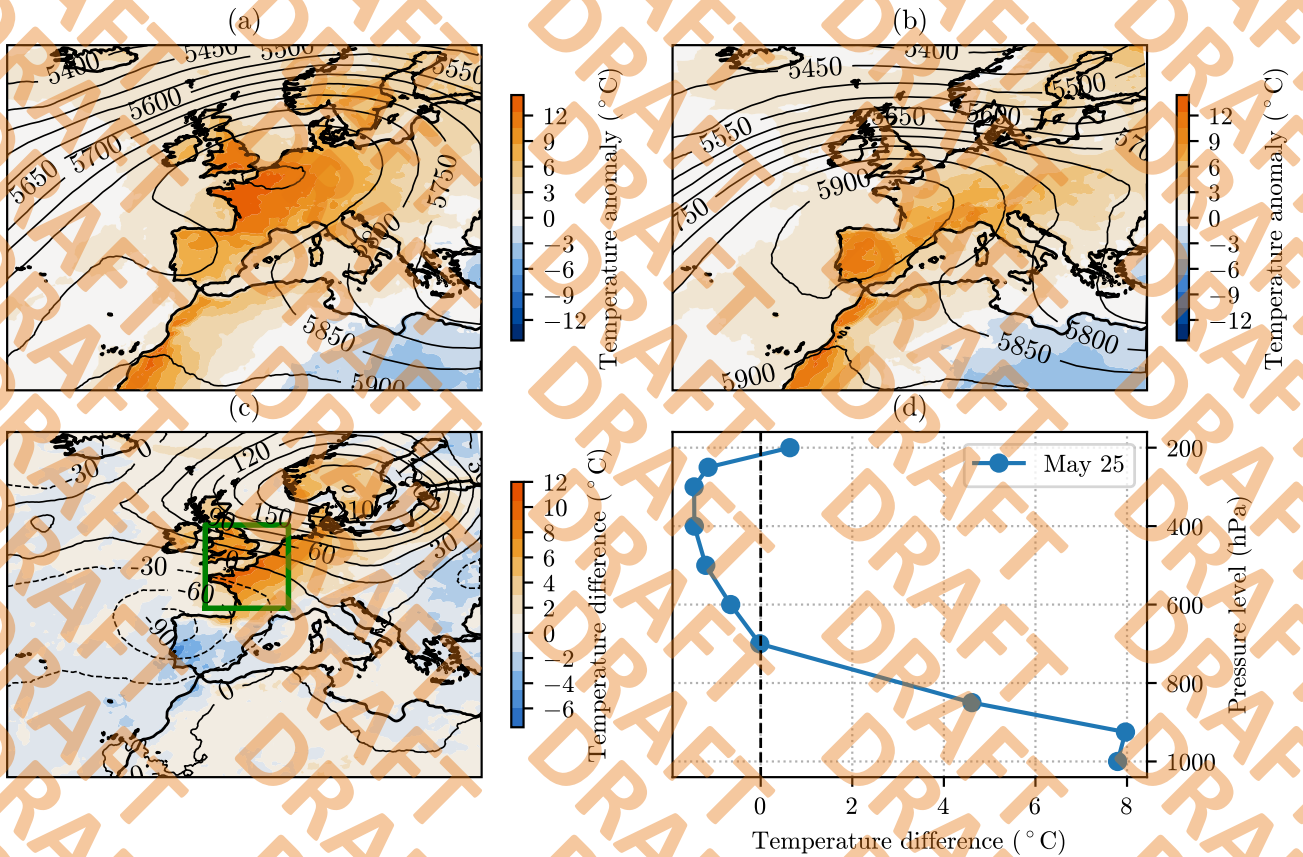


Figure 10 : Diagnostic of AIFS simulations. Of all members exceeding the threshold ($z_{500} > 5683\text{m}$), the 10 with the highest (a) and coldest (b) 12UTC near-surface temperature are selected, their temperature anomalies and 500 hPa geopotential are shown. (c) Composite difference of ENS1 and ENS2. (d) Difference of vertical temperature profiles between ENS1 and ENS2.

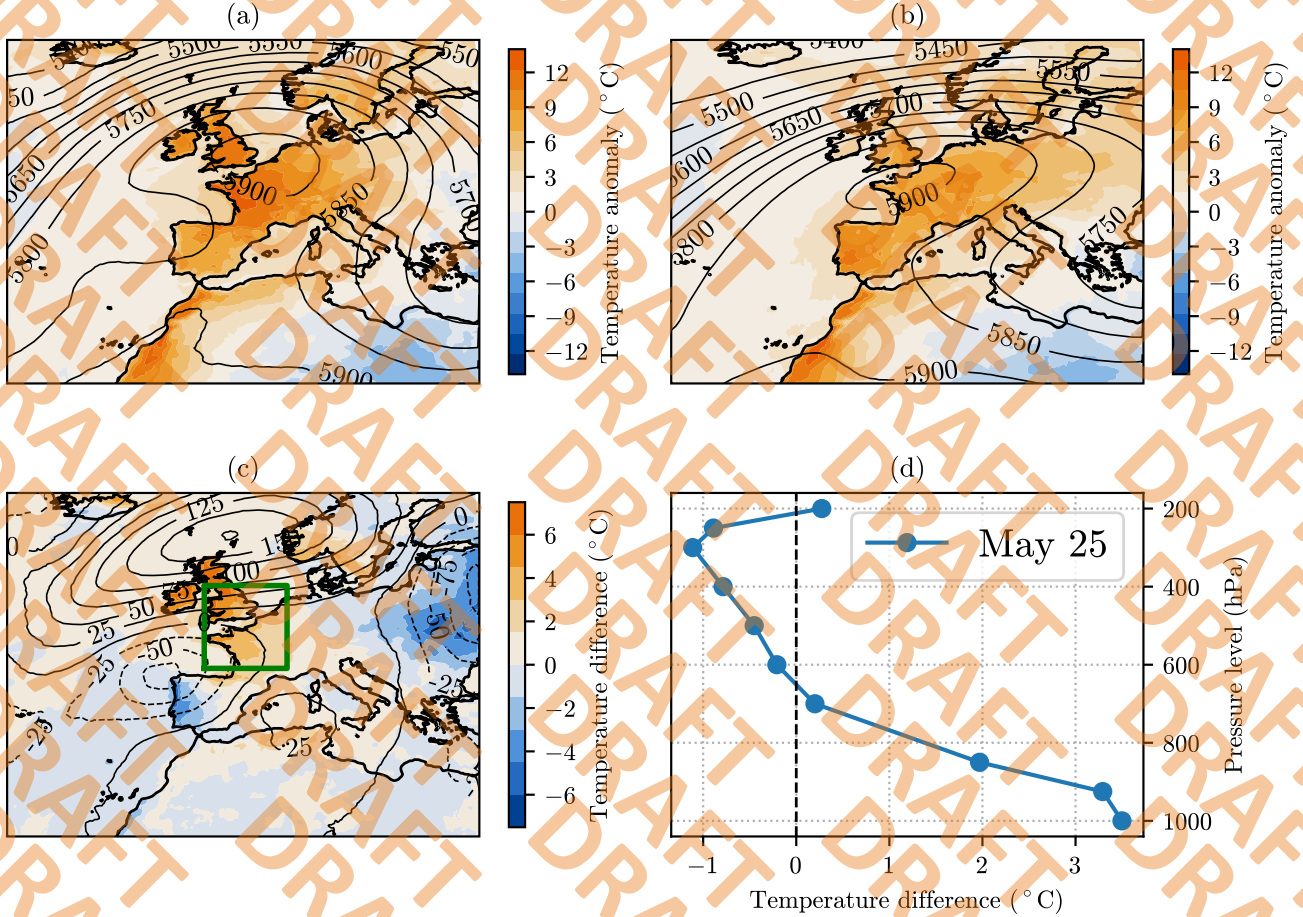


Figure 11 : Diagnostic of AIFS simulations. Of all members exceeding the threshold ($z_{500} > 5863\text{m}$), the 10 with the most intense cutoff according to (Kasuga et al., 2021) (a) and the rest (b) are selected, their temperature anomalies and 500 hPa geopotential are shown. (c) Composite difference of ENS1 and ENS2. (d) Difference of vertical temperature profiles between ENS1 and ENS2.

Africa. In the upper troposphere, the air is colder in the ENS1 members compared to ENS2, which likely is linked to the presence of the cut-off. This first comparison indicates that high pressures only are not enough to heat up the surface through an adiabatic process, and require an advection process to lead to such high surface temperatures.

To verify this, we select the top 10 members for each initialization date with the most intense cut-off low located in the region 37°N to 45°N and 17°W 2.5°W , with the methodology from (Kasuga et al., 2021). We obtain a third ensemble (ENS3). We also select the rest of the members without a cut-off low (ENS4) that still reach the 5683m threshold for z_{500} on the target area for a better comparison.

Comparing ENS3 with ENS4 (Figure 11) we make very similar observations as when comparing (ENS1) against (ENS2), though with less pronounced response which is to be expected given how (ENS1) and (ENS2) were selected. We also observe lower temperatures in (ENS3) against (ENS4) over the region where the cutoff low appears, this is to be expected as well given that they are westward cutoffs and bring Atlantic air in their center, this happens in (ENS1) against (ENS2) as well. The greatest temperature difference in the target area between (ENS1, ENS3) and (ENS2, ENS4) takes place at the surface and lower troposphere, indicating the strong role of advection rather than just subsidence in the context of this heatwave. This is confirmed in the decomposition of temperature (Figure 12, Section 2.5.3). ENS1 and ENS3 members are colder higher up in the atmosphere; this is investigated further using

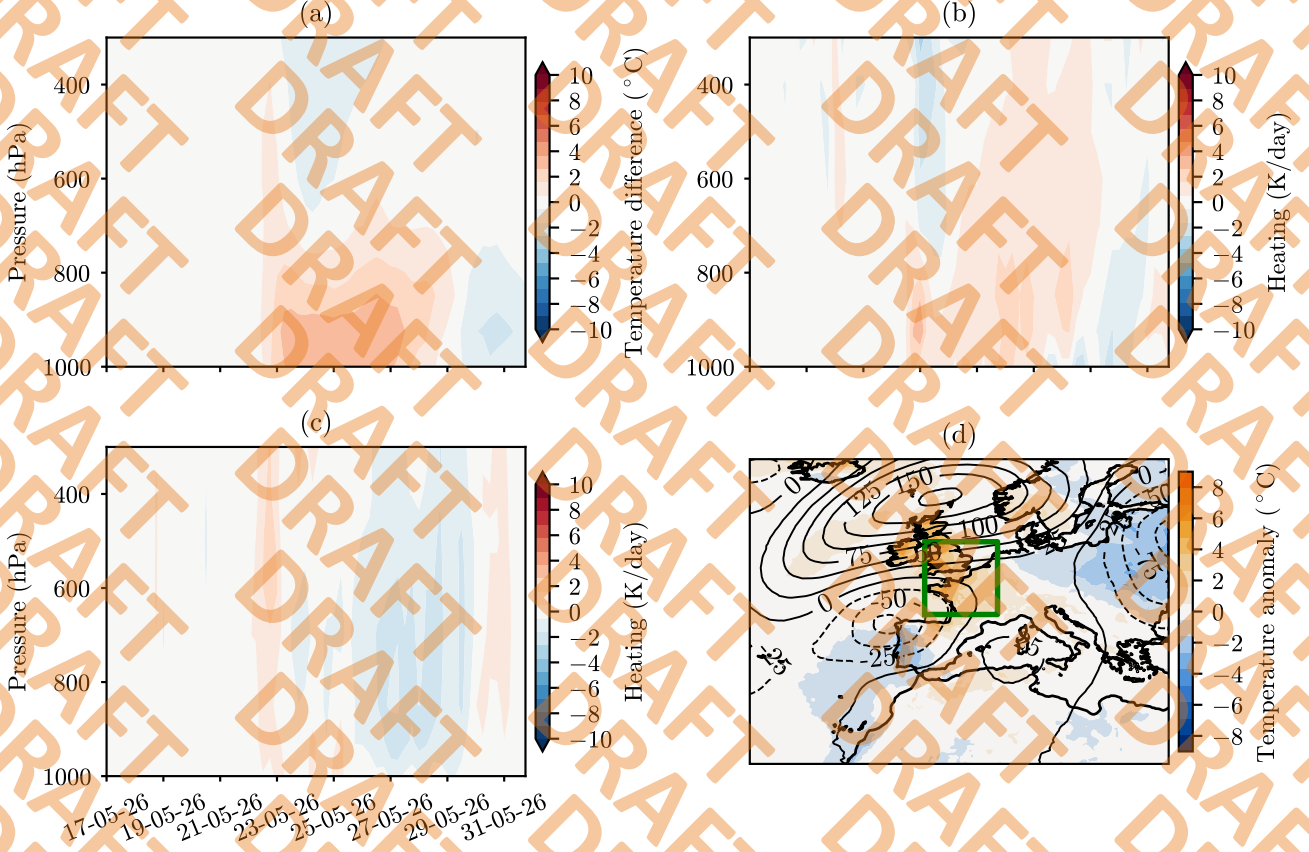


Figure 12 : Temperature decomposition of ENS3-ENS4. Spatially averaged over the green box. (a) The composite temperature difference. (b) The composite difference of advection terms $-V_h \cdot \nabla_h T$. (c) The composite difference of adiabatic heating $\omega(\frac{\kappa T}{p} - \frac{\partial T}{\partial p})$. (d) The fields composite differences on the 25th.

Lagrangian analysis in the paper to determine where the air parcels at different pressure levels come from. Nonetheless, this analysis of ENS3 versus ENS4 shows the occurrence of high temperatures in the case of this heatwave is driven by the combination of two synoptic patterns; a record breaking blocking and a cutoff low west of the Iberian peninsula.

As a sanity check, we also check that there is a significant difference between selecting from (ENS3) against selecting randomly through all members using bootstrap (Figure 13). We find that there is a significant difference in temperature around the whole target area. Across the significantly impacted grid-points in the target area, the presence of the cutoff low exacerbated the heatwave by an additional 1.7°C (init 18th) to 1.8°C (init 17th). The top 5% most severely affected grid-points experienced excess warming of 3.3°C (init 18th) to 3.5°C (init 17th) solely due to the presence of the cut-off, peaking at over 4.2°C excess in the most impacted grid-points.

We use the ensemble boosting estimator to obtain a better estimate of the return levels of extreme temperatures at 12UTC in May. The parameters used are $N_{\text{parent}} = 9$, $T_{\text{ref}} = 29.52^\circ\text{C}$, $N_{\text{lead}} = 3$ and $N_{\text{batch}} = 201$. This gives a total amount of boosted trajectories $N_b = N_{\text{lead}} \cdot N_{\text{batch}} \cdot N_{\text{parent}} = 5427$. The May 2026 WE heatwave was the most extreme in ERA5 in terms of intensity. After removing a long term trend to be able to compare events from different decades without the effect of climate change, it stands at the 4th place. We also fitted a Generalized Extreme Value (GEV) distribution to the May maximal temperature anomalies and we estimate a return period of 28 years event (Figure 14) for the May 2026 anomaly with Ensemble boosting and around 67 years with the GEV. From this analysis, a centennial

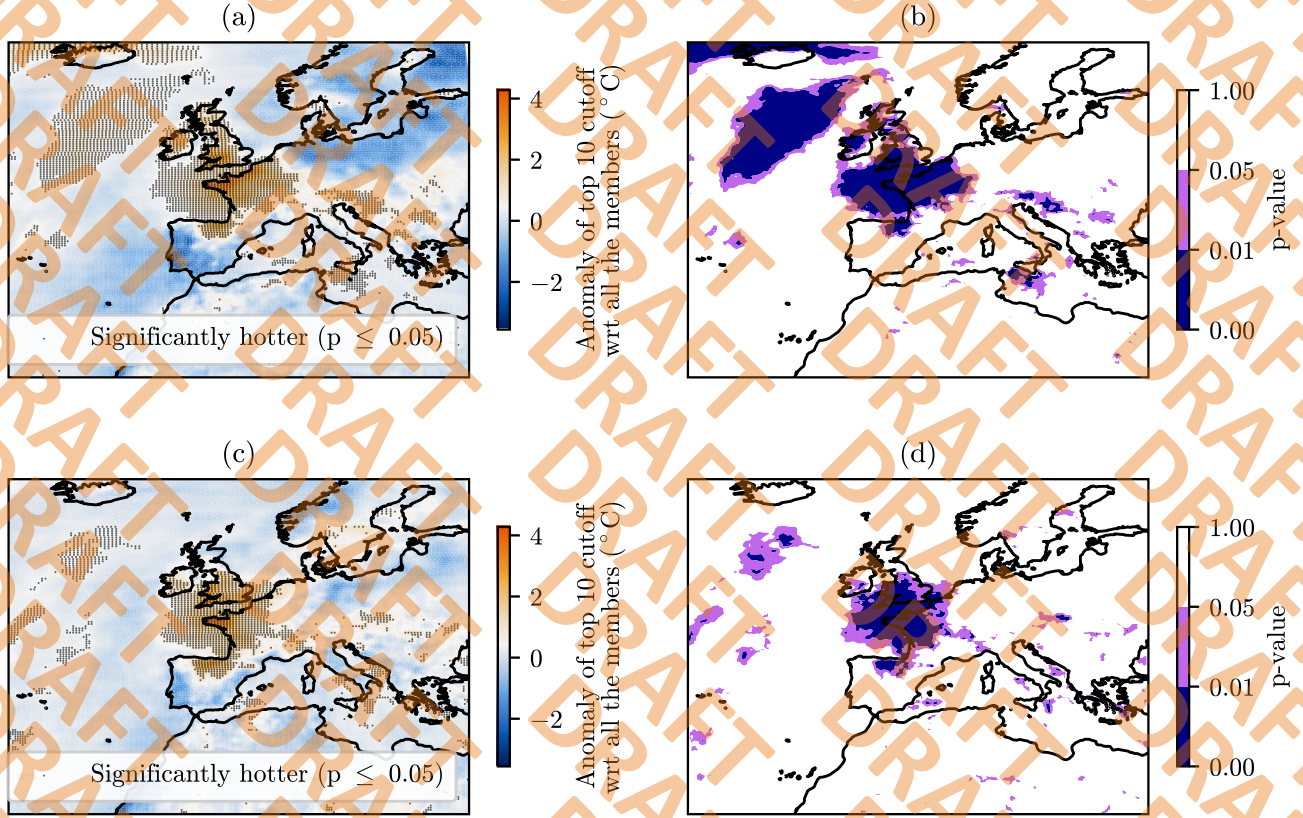


Figure 13 : Statistical significance of the cutoff low's impact on surface temperatures. Ensemble initialized on May 17th (top row) and May 18th (bottom row). (a,c) Daily mean 2-meter temperature anomaly ($^{\circ}\text{C}$) of the 10 ensemble members with the most intense cutoff lows relative to the full ensemble mean. Black points indicates regions that are significantly hotter ($p \leq 0.05$). (b,d) Spatial distribution of p -values. Statistical significance was calculated using a one-tailed bootstrap permutation test (2,000 iterations), comparing the 10-member cutoff composite against random 10-member subsets drawn from the full ensemble.

heatwave could reach temperatures 1.9°C higher than May 2026, and a millennial heatwave could be 4°C higher. Accounting for the typical $\sim 3^{\circ}\text{C}$ difference between ERA5 temperature at 12:00UTC grid-cell value and local station maxima, this translates to localized daily maximum temperature (T_{max}) of around 40°C . The estimations also indicates the GEV fit underestimates the temperature upper bound. Such a centennial or millennial heat wave would have devastating impacts. Early-season temperatures have increased impacts on public health and biodiversity as the bodies are not acclimated yet to high temperatures (Lee et al., 2014). Early-season heatwaves might also participate in drying soils through evaporation before the season even begins. Low soil moisture limits evapotranspiration, creating a positive land-atmosphere feedback loop, drying the atmosphere and therefore resulting in less clouds and precipitations (Miralles et al., 2019; Stéfanon et al., 2014). This increases the risks of extreme heatwaves and fires during the following summer.

When looking into a centennial event with temperature at 12UTC reaching 34°C for the Paris gridpoint (Figure 15), we observe a strong moist static energy imbalance between the surface and the tropopause (Noyelle et al., 2023b; Zhang & Boos, 2023). Therefore, the surface temperatures are close to the theoretical maximum allowed by soil moisture and available solar input for the given mid-tropospheric temperatures. Interestingly, in this case the instability is really consistent and temperature keeps growing after the first instability. This effect is mitigated when averaging horizontally over a bigger region though.

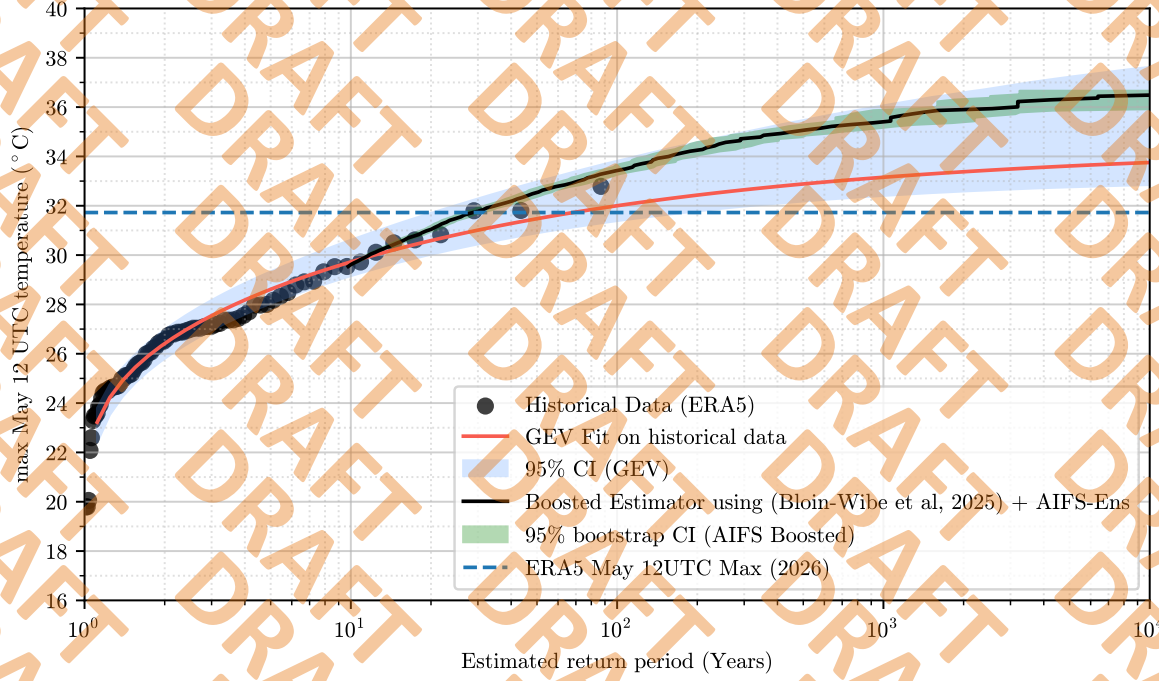


Figure 14 : Estimated return periods with the boosting estimator and a GEV.

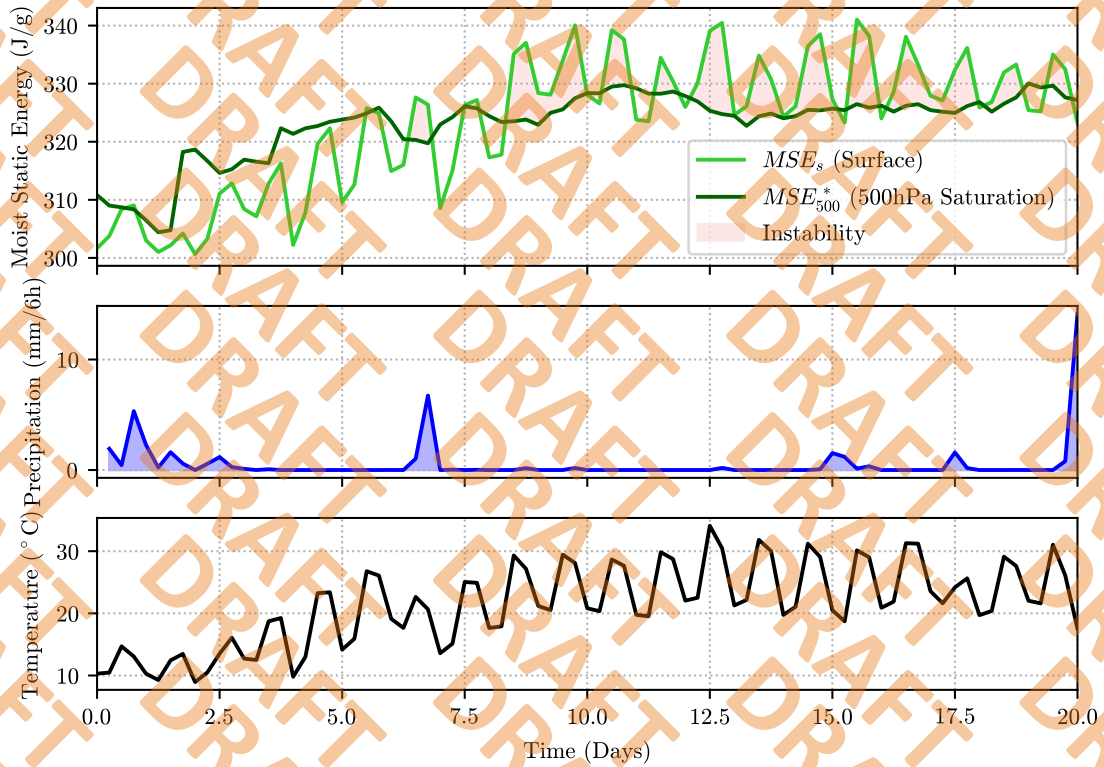


Figure 15 : Moist Static Energy instability of hottest member initialized on May, 17th at the Paris grid-point. (Top) Time series of surface moist static energy (MSE_s , light green line) and 500 hPa saturation moist static energy (MSE_{500}^* , dark green line) in J/g. Pink shaded areas highlight periods of convective instability ($MSE_s > MSE_{500}^*$). **(Middle)** 6-hourly precipitation accumulation (mm/6h). **(Bottom)** Surface temperature (°C).

In short, the May 2026 heatwave appears to be a very similar event to typical summer heatwaves, but that has found its way to May. We showed this heatwave was primarily driven by a blocking and amplified by a strong westerly cut-off low. The effect of climate change on this specific event will be studied in the paper.

3.2 Iterative boosting for summer maximum temperatures

In this second section, I will summarize here the results obtained with Iterative Boosting. The goal is to assess whether the method is capable of both producing extreme unseen temperatures and robust estimations of return times and levels.

We start by selecting $N_{\text{parent}} = 16$ events from the detrended summer temperatures. The iterative boosting is run two times with 480 members and $J = 4$ (R1) then $J = 3$ (R3) phases. We use $t_i = [11/12/13 \ 10 \ 5 \ 5]$ days and $f_i = [15 \ 10 \ 10 \ 10]$ days for R1 and $t_i = [11/12/13 \ 10 \ 5]$ days and $f_i = [15 \ 10 \ 10]$ days for R2. The threshold for these events $T_{\text{ref},0}$ is of 37.17°C while the ERA5 maximum is 39.8°C in July 2019 (40.42°C detrended). Note that for this methodology, I did not detrend the outputs, but assume they stay comparable with the detrended values used in the GEV¹⁸.

The results obtained in the two experiments are summarized in Figure 16 and shown in Figure 17. The R1 experiment yielded higher temperatures than R2 but its last phase was unsuccessful, in the sense that the maximum temperature of phase 4 was smaller than the penultimate threshold. It is therefore not possible to use this phase for any estimation and we discard it completely. Setting aside this stepback, the results show the algorithm is highly capable of reaching record breaking temperature with only a small amount of trajectories.

In Figure 17, we observe that only a few selected initial parent events seem to have been selected by the parent algorithm. Indeed, for R1 80% of events are initiated from 2022 event and the other 20% from the 2015 event, with similar results for R2 (respectively 67% and 33%). This is coherent according to the typicality argument from (Noyelle et al., 2023a), the more extreme the event the more typical the atmospheric situation. But in that case one could wonder why the historical maximum of 2019 did not yield good enough results. More trajectories might help with this, as 480 trajectories in total is possibly not enough to give every initial event a good chance at being selected.

(a) R1

Phase	Forward	Lead	Max Temp	Threshold ($q = 5.0\%$)
1	15 days	11,12 or 13 days	p1_m180 (40.84°C)	36.96°C
2	10 days	5 days	p2_m318 (43.21°C)	41.25°C
3	10 days	5 days	p3_m005 (44.90°C)	43.37°C
4	10 days	5 days	p4_m089 (41.56°C)	39.02°C

(b) R2

Phase	Forward	Lead	Max Temp	Threshold ($q = 5.0\%$)
1	15 days	11, 12 or 13 days	p1_m271 (42.38°C)	37.16°C
2	10 days	10 days	p2_m009 (41.55°C)	39.77°C
3	15 days	5 days	p3_m013 (43.13°C)	41.29°C

Figure 16 : Results from experiments R1 and R2 of iterative boosting.

¹⁸Results were not significantly different when the outputs of the ensemble boosting were detrended.

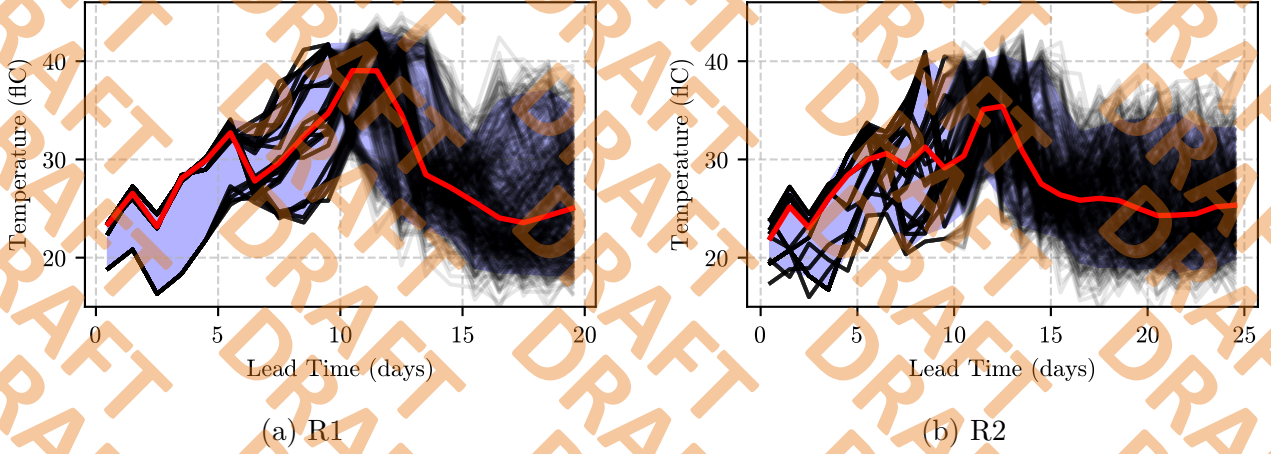


Figure 17 : Results from experiments R1 (a) and R2 (b) of iterative boosting. The reconstructed trajectories are represented in black, while the blue background fill is the 5%-95% range and the red line are the ensemble medians.

We analyze the R1 runs and find only one distinct atmospheric situation in the top 10 members. This situation is similar the one described in Section 3.1, which is a strong omega blocking over the whole of western Europe that is amplified by either a really deep through (Figure 18a, 1 out of 10) or a cutoff low (Figure 18b) west of the Iberian Peninsula (9 out of 10). The temperature decompositions for these events (Figure 19) shows the anomalies seem driven mostly by the adiabatic term, because of the subsidence from the blocking and that the advection isn't enough to cool down the lower levels as it has no effect. The diabatic term we did not compute likely also plays an important role due to the clear skies that are associated with this type of synoptic patterns.

We now use the R2 runs to estimate return times using the methodology proposed in Section 2.4.2 to compare it with return times estimations obtained with a GEV on the detrended temperatures (Figure 20). The estimation we obtain shows the GEV slightly underestimates the return levels and is less confident for longer return periods.

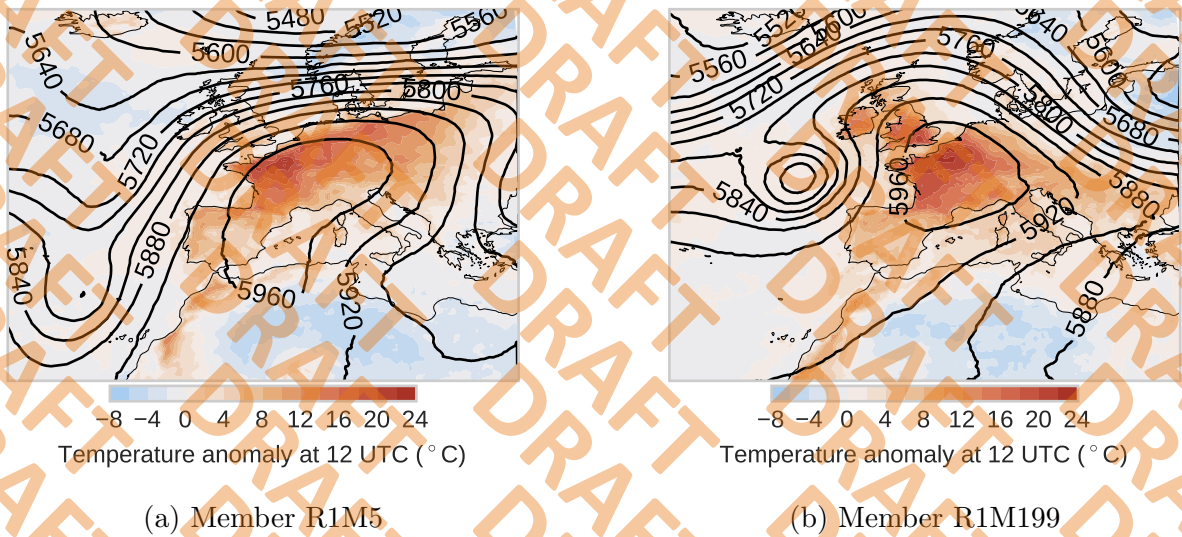


Figure 18 : Temperature anomalies at 12UTC and daily averaged geopotential maps for the warmest events in R1. The isolines represent the geopotential. (a) Is the hottest event while (b) is the second hottest. The 8 other members in the top 10 exhibit a cutoff low like (b) and look fairly similar.

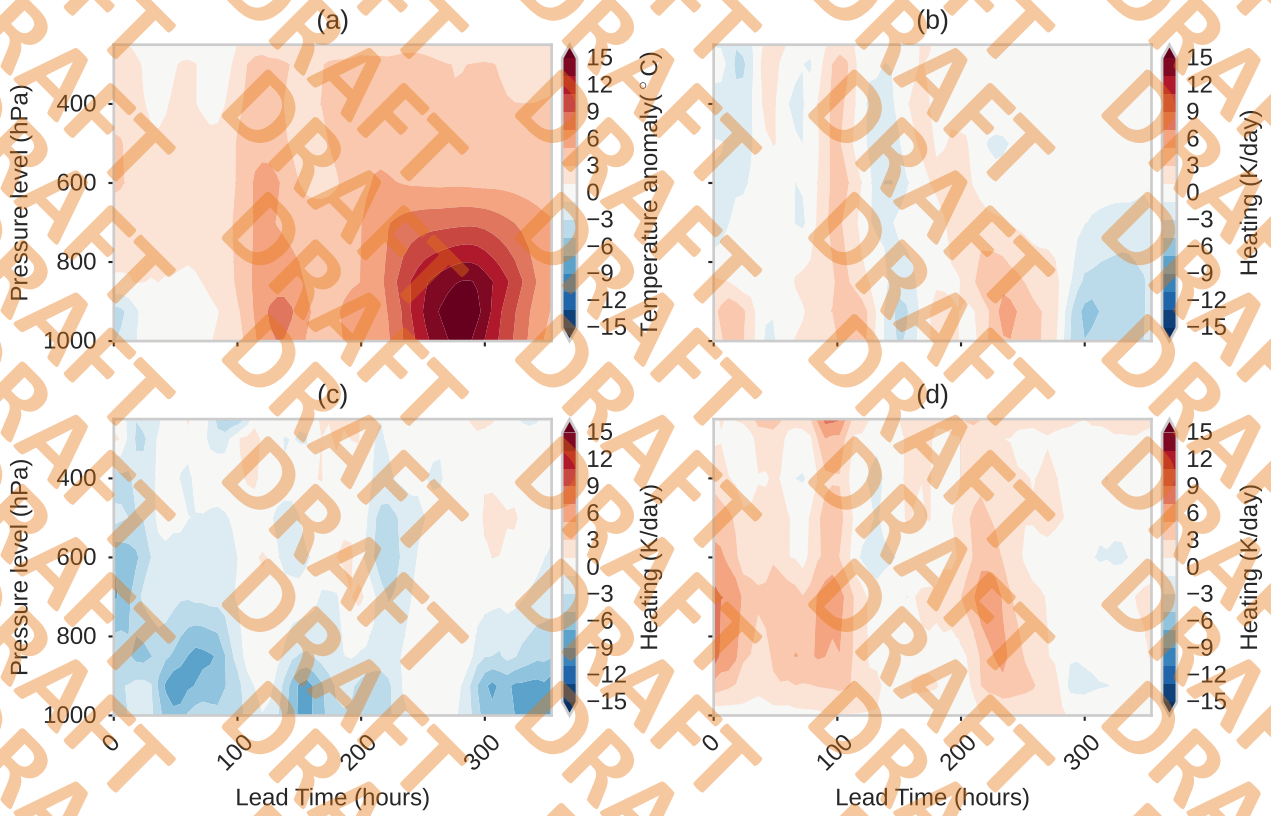


Figure 19 : Temperature decomposition of the 10 hottest R1 members. Spatially averaged over the Ile de France region. (a) The temperature anomaly. (b) The composites of rate of change of temperature $\frac{\partial T}{\partial t}$. (c) The composites of the advection terms $-V_h \cdot \nabla_h T$. (d) The composites of the adiabatic heating $\omega\left(\frac{\kappa T}{p} - \frac{\partial T}{\partial p}\right)$.

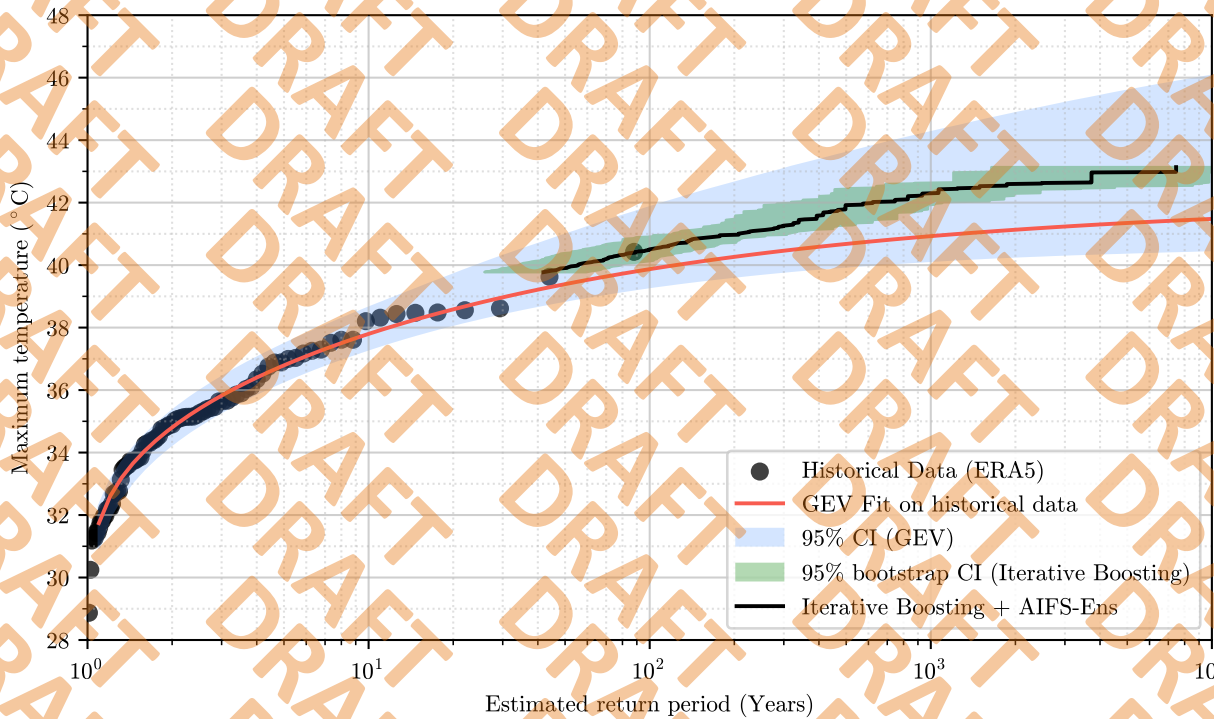


Figure 20 : Estimated return periods with the iterative boosting estimator and a GEV.

To summarize, the iterative boosting algorithm proved very efficient in reaching extreme events at a reduced cost. It can be used for studying very extreme events at the fraction of the cost of running long climate simulations.

DISCUSSIONS

We managed to use AIFS in conjunction with rare events algorithms but the methodology we propose has some limitations.

The main limitation of this work is the model itself. As explained in Sup D-b-1, it is not possible to use AIFS-Ens for simulations longer than 25 to 30 days. This limits for now the uses of rare event algorithms with AI models to boosting methods, as to the best of my knowledge there are no other methods usable that do not require longer time integration in the likes of GKLT (Ragone et al., 2018). It is with no doubt, that future models will be usable in more situations. That being said, these AI models are remarkably good at “resolving” physics and conserving good physical and statistical properties.

Another important fact to note is that, for our methodologies, we use data that is present in the training distribution of AIFS-Ens. This is mitigated by the fact that the model is non deterministic and was fine-tuned on IFS ensemble predictions. Using the same methodologies with a deterministic model, would be questionable for that exact reason unless maybe one uses sophisticated initial perturbation schemes like BRED vectors (Toth & Kalnay, 1993). This might also explain why the deterministic AIFS-Single was inconclusive in showing good statistical properties (Sup D), but on the other hand one of our test cases was out of training distribution. Nevertheless, probabilistic models are inherently more realistic as they natively represent the uncertainties associated with the climate system.

The third point to be discussed is whether the outputs for both ensemble boosting methods are to be detrended. Indeed one could argue we assume the climate to be stationary so the detrending is necessary, similarly to when fitting a GEV. But there is no certainty that AIFS-Ens is able to capture long term trends, meaning we do not know if we can assume all outputs to be from the current climate or the climate they were initially sampled from. When I did some tests for the ensemble boosting of May temperatures I did not find significant differences between both methods. But, it is important to note that if one assumes that AIFS-Ens outputs current climate predictions, that the selection of initial parent events and the computing of the baseline probability should still be done with respect to the detrended temperatures, as I did in Section 3.2. This should be further tested.

The work on the 2026 heatwave was novel in our approach of ensemble boosting with AI models to generate counterfactual simulations, and yielded interesting results. Though these results still need to be backed up by a Lagrangian analysis – where one follows the air parcel to find their origin and to estimate a more sophisticated decomposition of the temperature anomaly – which is addressed in the article.

Iterative boosting, while extremely efficient, still lacks some robustness when used paired with AIFS-Ens. This could however rather easily be fixed by properly parallelizing the algorithm for AIFS-Ens which would allow for more trajectories and therefore more robust estimations. That being said, the estimations we get are already really good, but the differences in maximum temperatures and thresholds between our experiments shows there is a need for more trajectories. For comparison, each of our experiments used 480 trajectories in total with similar integration time as for ensemble boosting in Section 3.1. The ensemble boosting methodology we applied used however a total of 5427 trajectories. We reach orders of magnitude comparable with 11 times less trajectories with iterative boosting. Parallelizing such that for example each selected parent’s trajectories are run on an independant GPU would allow for bigger simulations and more robust results. This is especially true for the estimation of return times, as the

results in obtained with the two simulations R1 (Figure 16) and R2 (Figure 20) were fairly different, indicating a strong sensitivity to the parameters of iterative boosting. The very tail of the distribution seemed well estimated in both cases, but the higher temperatures in R2 seemed to have shifted the return levels higher, more members should fix this and render it more robust. The method however seems very promising.

CONCLUSION

The goal of this internship was simple, assess the possibility of using rare events algorithms with AIFS. For that, after having validated the physical and statistical behaviour of ECMWF's AIFS-Ens model, I used it to perform rare event algorithms, including a case study of the May 2026 WE heatwave. Finally, I went further than the initial objective and presented and evaluated a new rare event algorithm method, Iterative Boosting. Which is an extension of the regular Ensemble Boosting thought for AI models, models which for now require shorter integration times than physical models to stay consistent.

To answer the initial question, I showed AIFS-Ens is very capable for rare events algorithms. It showed great physical consistency and statistical properties, including a chaotic behaviour. This leads me to conclusion on both the May 2026 WE Heatwave case study and iterative boosting.

For the May 2026 Heatwave, we showed through ensemble boosting that a Cut-off low over Portugal and Spain significantly amplified daily mean surface temperatures in southern UK and western France by up to 4.3°C locally. This cut-off enhanced the primary heatwave drivers of adiabatic and advective warming at the surface. Ensemble Boosting gives an estimate of 40°C in Paris for a millennial type of May heatwave in the current climate, making it comparable to the June 2026 Heatwave and providing an adaption mark for early-season heatwaves.

Concerning iterative boosting, I showed the algorithm is highly efficient on a simple Ornstein Uhlenbeck process, even when compared with already efficient simple ensemble boosting. The algorithm also proved very capable in a more realistic use case, enabling us to simulate heatwaves 5 degrees warmer than the historical maximum in ERA5. This could be really interesting to assess the impacts of such a heatwave and help better prepare infrastructures or systems to what is possible under current climate, via more precise estimations of return times compared to fitting GEV distributions. This is crucial for public health and better advisement of policymakers, as a 45°C heatwave at 12UTC on ERA5 grid could mean up to 48°C locally if we account for the usual difference with temperature locally measured at weather stations.

Other future perspectives, outside of adresssing what was brought up in the last section could be to work with pre-existing long climate simulation such as the ones made with IPSL-CM6A-LR (Boucher et al., 2020). See how the model adapts to future climate to perhaps boost hypothetical future events. If this were to be inconclusive or impossible due to inputs mismatch, another possibility could be to train a smaller version of AIFS on IPSL-CM6A-LR simulations. These simulations have coarser grid, this would allow one to have an emulator of the climate model and use it to perform rare event algorithms in future climate. The latter solution would bring its own set of issues that would have to be addressed. Starting with the bias, learning from physical climate models would imply learning the inherent bias of said climate model.

This work opened the door for using AIFS-Ens and rare events algorithms in conjunction, I am sure it will be of use to other students or researchers. There are still a lot of questions left unanswered and perspectives to be explored. The part about the May 2026 heatwave will be included in an article from Maeve Mayer in which I will be second author. The work about iterative boosting could be the subject of another article, if some of the discussions in the last section are adresssed, but this report could serve

as a basis. All the code produced was thought to be readable and understandable by another user, while being optimized.

In terms of personal assesment, I would say this internship was a formative and enriching experience further motivating me to continue working in applied maths for climate and environmental matters. I had the opportunity to meet a lot of passionate and interesting people. This internship allowed me work in a lab dedicated to to thematics I am sensible to, and I am grateful for having the chance to be able to choose such a path for myself as a future applied mathematics graduate. This internship also allowed me to continue developping my appetite for research and critical thinking. Not only did I strengthened my technical knowledge but also my capacity to deal with problems and difficulties in research.

Starting next year, I will be a PhD candidate at Université Paris Cité and LISA (Laboratoire Interuniversitaire des Systèmes Atmosphériques), funded by the CNES (Centre National d'Etudes Spatiales), where I will work on the estimation of pollutant emissions from space using deep learning. This will be another opportunity for me to work on matters that resonate with me, and an adventure in which I am sure the lessons I learnt from this internship will be of use.

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As per the GENCI documentation on carbon footprint, this work was responsible for 22kg of CO₂ equivalent. Note that this footprint, calculated using the Labos 1point5 method covers the IT equipment (computing, temporary storage, network) and its cooling system, but does not include building construction or personnel transportation, nor do I include the actual training of the two AIFS Models.

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CODE AVAILABILITY

The AIFS model is available on [Huggingface](#). All my code is available on the project's [github repository](#).

USE OF AI

In the context of this internship, AI (namely Google Gemini¹⁹ and Mistral's Emmy²⁰ (CNRS assistant)) were mostly used as an assistant, such as explaining parts of articles or help with some coding tasks or for start of a bibliography study when needed. No AI was used to generate text for this report and every information given was carefully verified with academic sources.

GLOSSARY

Advection Horizontal transport of the properties of a mass of air (temperature, humidity etc)

Cut-off low A cold-cored depression in the upper troposphere, that became detached or cut off from the main westerly atmospheric circulation (or jet stream).

Diabatic heating A process which occurs with the addition or loss of heat. The opposite of adiabatic. Meteorological examples include air parcels warming due to the absorption of infrared radiation or release of latent heat.

Subsidence A descending motion of air in the atmosphere, which causes adiabatic compression and warming of the air parcel.

AMS Adaptive Multilevel Splitting.

GEV Generalized Extreme Value distribution, used to model the occurrence of extreme observations

GNN Graph Neural Network

TAMS Trajectory-Adaptive Multilevel Splitting.

TEAMS Trying-Early Adaptive Multilevel Splitting.

AIFS Artificial Intelligence Forecasting System (ECMWF's data-driven model).

ECMWF European Centre for Medium-Range Weather Forecasts.

ERA5 5th generation ECMWF atmospheric reanalysis of the global climate.

IFS Integrated Forecasting System (ECMWF's traditional, physics-based numerical weather prediction system).

LSCE Laboratoire des Sciences du Climat et de l'Environnement.

NWP Numerical Weather Prediction.

¹⁹Google. (2026) Gemini (3.1 Pro) [LLM]

²⁰Mistral. (2026) Emmy (CNRS Version) [LLM]



SUPPLEMENTARY MATERIAL

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A GROWTH RATES IN IFS

B CRPS LOSS

We broadly introduce here the Continuous Ranked Probability Score (CRPS) loss and its various extensions. The CRPS was introduced in the 70s (Brown, 1974) and improved by (Hersbach, 2000).

Definition 2.1 (CRPS). Noting F as the predictive CDF and y as the verifying observation of a variable of interest.

$$\text{CRPS}(F, y) = \int_{-\infty}^{+\infty} [F(t) - \mathbb{1}(t \geq y)]^2 dt \quad (49)$$

The CRPS is a generalisation of the MAE for probabilistic forecasting, indeed, if the prediction is deterministic it is easy to see it reduces to the MAE. The CRPS has other nice properties including one on its decomposition.

Property 2.2 (Decomposition of the CRPS). It has been showed (Gneiting et al., 2005) that the CRPS can be written in terms of expectations when given multiple forecasts.

$$\text{CRPS}(F, y) = \mathbb{E}[|X - y|] - \frac{1}{2} \mathbb{E}[|X - X^*|] \quad (50)$$

Where X and X^* are independent random variables associated to the forecast distribution F .

This property leads to the formulations in (Lang et al., 2026).





Figure S1 : Plot of all the pairwise log differences between 50 ensemble trajectories starting from a same day in IFS computed on the 500hPa geopotential.



Definition 2.3 (CRPS for ensemble forecasting). Given an M-member forecast ensemble with members $\{x_j\}_{j=1,\dots,M}$ and y the verifying observation, the CRPS writes as

$$\text{CRPS}\left(\{x_j\}_{j=1}^M, y\right) = \frac{1}{M} \sum_{j=1}^M |x_j - y| - \frac{1}{2M^2} \sum_{j=1}^M \sum_{k=1}^M |x_j - x_k|. \quad (51)$$

In order to account for ensemble size and penalize ensembles whose members do not behave as if they and the verifying observation were sampled from the same distribution we can introduce the fair CRPS (Ferro, 2014).

$$\text{fCRPS}\left(\{x_j\}_{j=1}^M, y\right) = \frac{1}{M} \sum_{j=1}^M |x_j - y| - \frac{1}{2M(M-1)} \sum_{j=1}^M \sum_{k=1}^M |x_j - x_k| \quad (52)$$

Then, because machine learned models are often trained on mixed precision for computational efficiency. It is crucial to avoid degeneracy (in the case where all members apart from one have the same value as the verifying observation, the last member is unconstrained) by introducing the almost fair CRPS where α is an hyperparameter.

$$\text{afCRPS}\left(\{x_j\}_{j=1}^M, y\right) = \alpha \times \text{fCRPS} + (1 - \alpha) \times \text{CRPS} \quad (53)$$

However, CRPS scores are scale unaware, they are only point-wise scores. This issue was tackled by (Lang et al., 2025) where a multi-scale loss formulation of the CRPS was proposed by introducing a sequence of ordered smoothing operators, each targeting different scales.

C THE GENERATION OF PERTURBATIONS

To generate new trajectories we take some initial conditions of the climate system $X^{(0)}$ and apply some perturbation to it. Using i as the index of the variable we want to apply the perturbation and applying it to the all the grid points. The same perturbation is added to both initial timestamps.

- Uniform perturbations : $\tilde{X}_i^{(0)} = X_i^{(0)} (1 + \sigma \times U([-1/2, 1/2]))$
- Gaussian perturbations : $\tilde{X}_i^{(0)} = X_i^{(0)} (1 + \mathcal{N}(0, \sigma))$
- Spatially 2D correlated noise : To generate spatially correlated noise we use a fractional Brownian motion (fBm) (Saupe, 1988) of the same shape as the inputs. It allows for the generation of fields like Figure 2. The transformation is $\tilde{X}_i^{(0)} = X_i^{(0)} (1 + \sigma \times \text{fBm})$.



Figure S2 : Example of a 2D spatially correlated noise using a Fourier filtering method.

When using AIFS-Ens however, there is no need to perturbate the initial conditions, as explained in Section 2.3.2.



D TESTS OF AIFS

For our tests, we will mostly be focusing on three specific heatwaves that each represent specific cases we are interested in. They are the following:

Date	Case
August 2003	Long lasting heatwave.
July 2019	Record temperature in Paris and other European cities.
June 2025	Heatwave that is not included in any model's training data.

D-a Preliminary results with AIFS Single

D-a-1 GPU induced chaos

AIFS being a model that runs on GPUs means it is subject to GPU determinism, which is limited in the sense that most algorithms used by deep learning libraries are actually non deterministic (Pham et al., 2020) and use mixed precision (Micikevicius et al., 2018). This means that even with two completely identical initial conditions, the model will never output the same results as seen in Figure 3b. This is not something that is usually impactful for deep learning frameworks but we suspect that the forecast being autoregressive might amplify this effect. Although this might look like an inconvenience, this is actually great or at least non inconvenient for our use cases as long as the simulations stay physically consistent. Which appears to be the case when observing the evolution of crucial fields like height geopotential (Figure S5).

In order to assess the system's response to varying initial conditions we need isolate the chaos generated by our perturbations and the one generated by the GPU. This is done through Pytorch options and significantly increases compute times by 4 but is necessary (Figure 3a) to isolate whether the divergence comes from the perturbations or the GPU.

D-a-2 “Small” Perturbation induced chaos

Multiple scales and variable to perturbate (see Sup C for details on how the fields were perturbed) were tried and are recapped in Table 1 with corresponding results.

From testing, we found that the field we perturb and the scale at which we do does not significantly changes the results (Table 1). Results which in all cases are deemed insufficient for rare events algorithms.

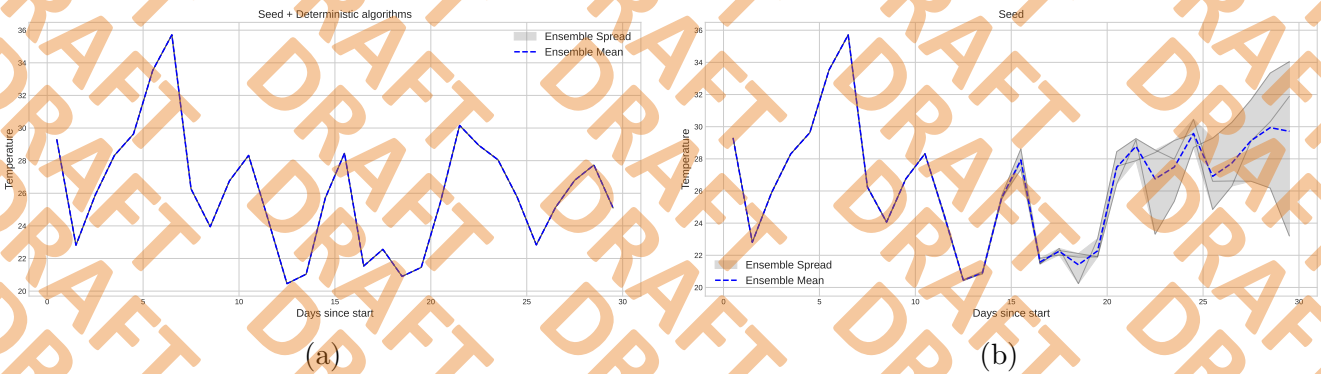


Figure S3 : Time series of temperatures for the Paris gridpoint generated using AIFS-Single from 3 exact same initial conditions. (a) We notice that after approximately 15 days the effects of non determinism on GPU start to generate chaos. (b) The exact same initial conditions do results in exact same trajectories.

Table S1 : Different experiments made to attempt to generate chaos from AIFS-Single ensemble forecasts. When multiple scales or perturbed variables are showed almost all or all combinations were tried. Cases where perturbations too big caused a period of convergence to the attractors were not taken into account. Growth rates are computed on 500hPa geopotential over the region 60°W to 60°E and 15°S to 70°N. When growth rates have been computed for multiple experiments, the interval is the interval in which all the single 95% bootstrap are.

Perturbation type	Scales	Perturbed variable	Start date	Growth rates	Time for divergence to start and be visible on the surface temperature (qualitative)
Gaussian	10^{-3} , 10^{-5} , 10^{-7} , 10^{-9}	2t, q, u+v, u+v+q	2025-06-25	[0.32, 0.4]	~ 15 to 20 days
Uniform	10^{-1} , 10^{-2} , 10^{-3} , 10^{-5} , 10^{-7} , 10^{-9}	2t, q, u+v, u+v+q	2025-06-25, 2021-06-14	[0.34, 0.39]	~ 15 to 20 days
fBm	10^{-1} , 10^{-3} , 10^{-5} , 10^{-7}	2t, q, u+v	2025-06-25	[0.23, 0.37]	~ 12 to 20 days
EDA from ERA5	Intrinsic	All	2025-06-25, 2021-06-14	[0.348, 0.368]	~ 10 days

These algorithms are not relevant if performed with a model that is not chaotic or takes too long (see Figure 4, Figure S5 and Figure S8 Figure S7) to show chaos.

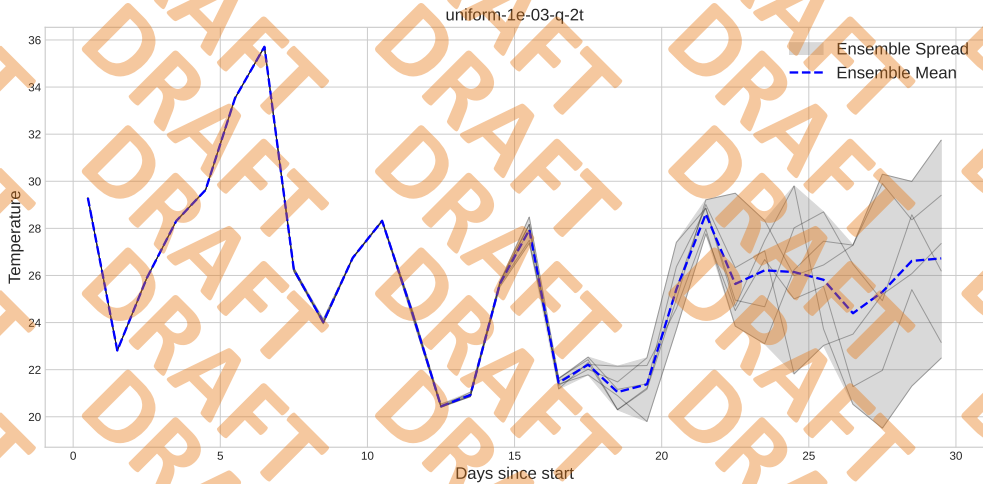


Figure S4 : An example of trajectories of temperatures at 12UTC time for Paris everyday that only diverge visibly after 15~20 days. The perturbation was done on the specific humidity q at a 10^{-3} scale.



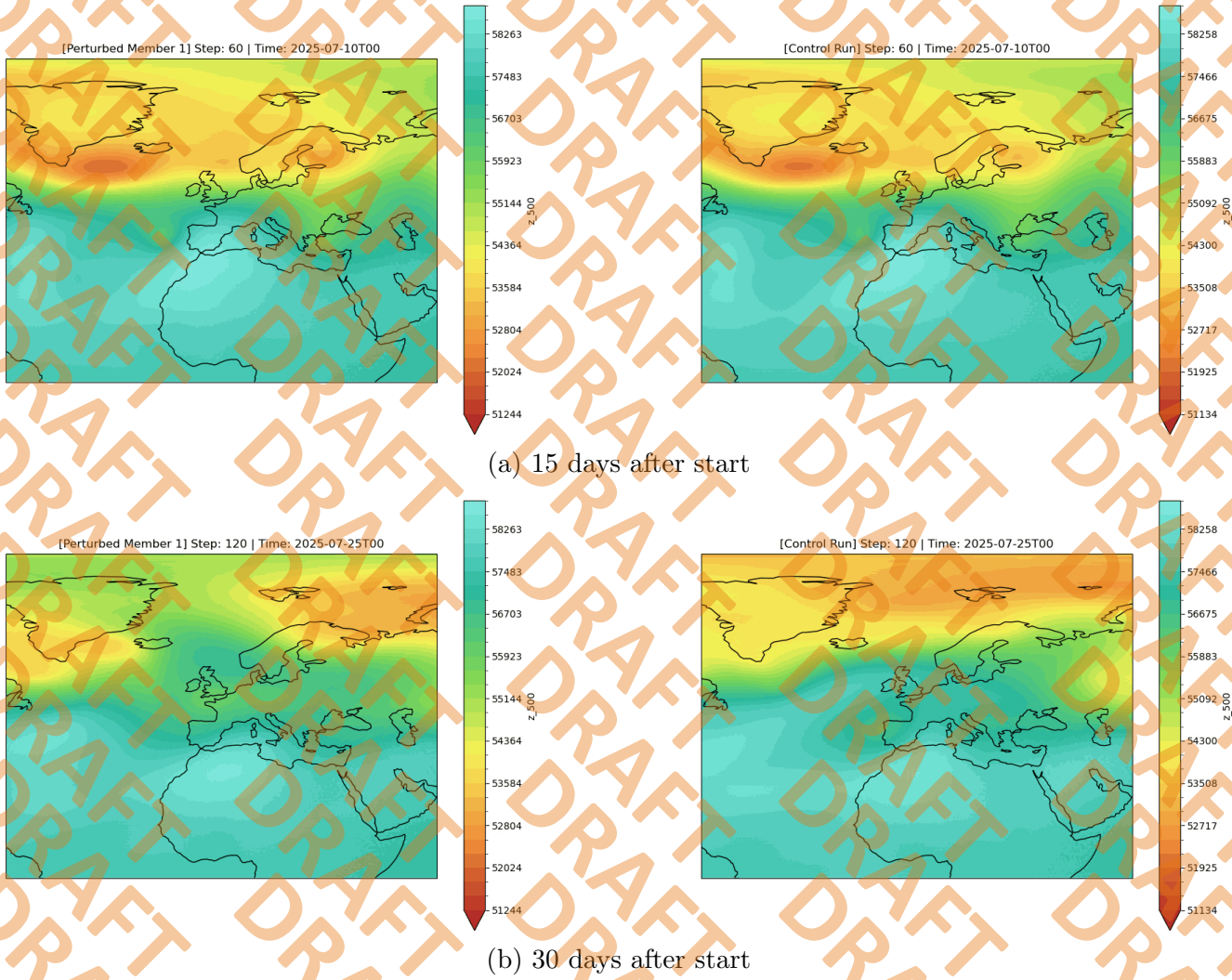


Figure S5 : Geopotential heights fields generated using AIFS-Single from 2 closely perturbed initial conditions. We notice that after approximately 15 days (a) the effects of the perturbations still haven't significantly changed the atmospheric configuration as compared with looking after 30 days (b) where we see a genuine difference between the two members.

The only occurrence where the scale of the perturbations yielded a fast enough significant change was when applying perturbations of scale 10^{-1} or 10^{-2} , which is ridiculously bigger than the numerical errors used in literature (represents approximately $\pm 13^{\circ}\text{C}$ and $\pm 1.3^{\circ}\text{C}$ respectively with our uniform perturbation if we are to perturb the 2m temperature). At this scale we observed a strange behaviour (visible on the gridpoint time series and pairwise log-distances of surface temperature Figure S6) of the model. It would dampen the initial perturbations, as if we put the system in a state far from the attractor and the simulation starts by converging to the attractor first.

These findings justify the change to injecting perturbations directly in the latent space and to a model trained for CRPS, namely AIFS-CRPS.



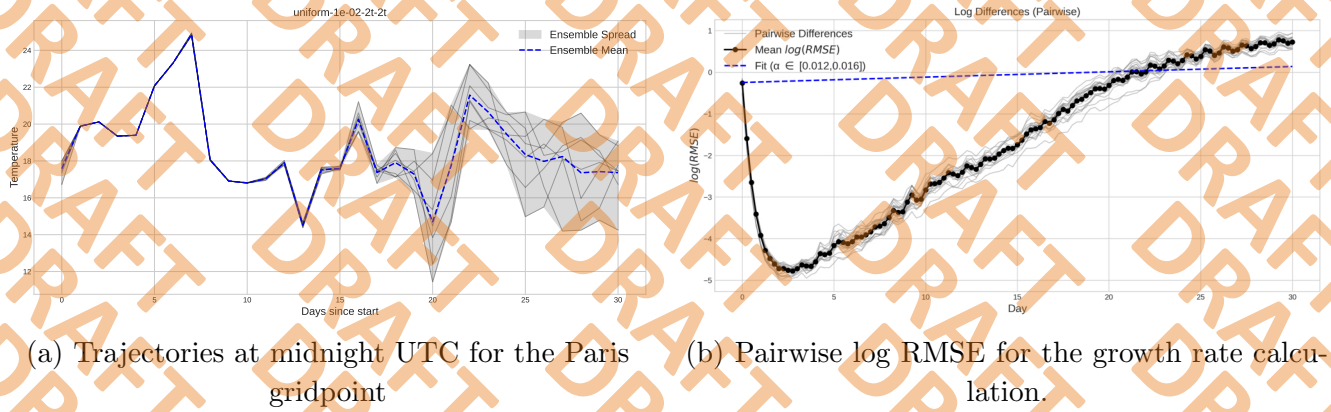


Figure S6 : Damping effect when the perturbation is too aggressive. The perturbation was done on the 2 meter surface temperature at a scale of 10^{-2} on the temperature. (a) The perturbation is erased by AIFS on the first 2 steps of the time series that. (b) The pairwise log-distances of temperatures start to diminish on the first step, which indicates that the system is aiming towards the attractor.

D-b Results with AIFS-CRPS

D-b-1 An important caveat regarding AIFS-Ens, unphysical artifacts for long rollouts

When trying to evaluate AIFS-CRPS on longer rollouts of about 90 days to assess whether we're able to simulate a full summer we stumbled onto an issue. Fields like geopotential or sea level pressure become completely unphysical after 40 days. The artefacts start from areas with high elevation after around 25 days and grow into worldwide anomalies. This is a known issue from what is reported on ECMWF's issue tracker²¹ but not to that extent. Indeed, the original article from ECMWF regarding the release of AIFS-CRPS (Lang et al., 2026) mentions they are able to perform subseasonal to seasonal (S2S) forecasts using AIFS-CRPS, or rather a version of CRPS running on lower 1° resolution 096 grid and O48 processor grid. We do not see why this would work with the 1° model and not with the higher resolution one. From discussions with Mariana Clare and Simon Lang from ECMWF, this is a known issue that should be fixed in the next version and stems from the fact that some models were fine-tuned on IFS operational data, for ease of integrating in ECMWF's system²². With that knowledge, we will not be running simulations for more than 25/30 days.

D-b-2 Latent space perturbations induced chaos

As explained in Section 2.3.2. It is easy to generate new independent trajectories from a same starting point with AIFS-CRPS thanks to the noise being added directly in the latent space. The trajectories (Figure S10) are clearly chaotic and we get growth rates (Figure S11) in the same range as with IFS or AIFS-Single. The difference with AIFS-Single, even though the growth rates are similar, is that the log-distances curves really exhibit a behaviour more in line with what is expected as explained in Section 2.2, that is we can observe the moment the trajectories start being completely uncorrelated. Therefore, the growth rates are more interpretable here. As a matter of fact, the formulation of the growth rate from Lorenz helps computing the growth rate or Lyapunov exponent for systems in which the plateau has not been reached yet so that explains why growth rates are similar. The divergence here happens way earlier after as little as 6-7 days. If we are to compute the growth rates on the 500hPa geopotential

²¹<https://confluence.ecmwf.int/display/FCST/Known+AIFS+ENS+Forecasting+Issues> topic G1

²²Moreover, we will not be using AIFS-Ens v2 as the performances of that model are degraded when using ERA5 data as initial conditions, in comparison with AIFS-Ens v1 whose performances stay great in that case.



Figure S7 : Time series of temperatures at 12UTC in Paris for different initial perturbations run with AIFS.





Figure S8 : Log distances for timestamps 12UTC calculated on the northern hemisphere for different initial perturbations run with AIFS.



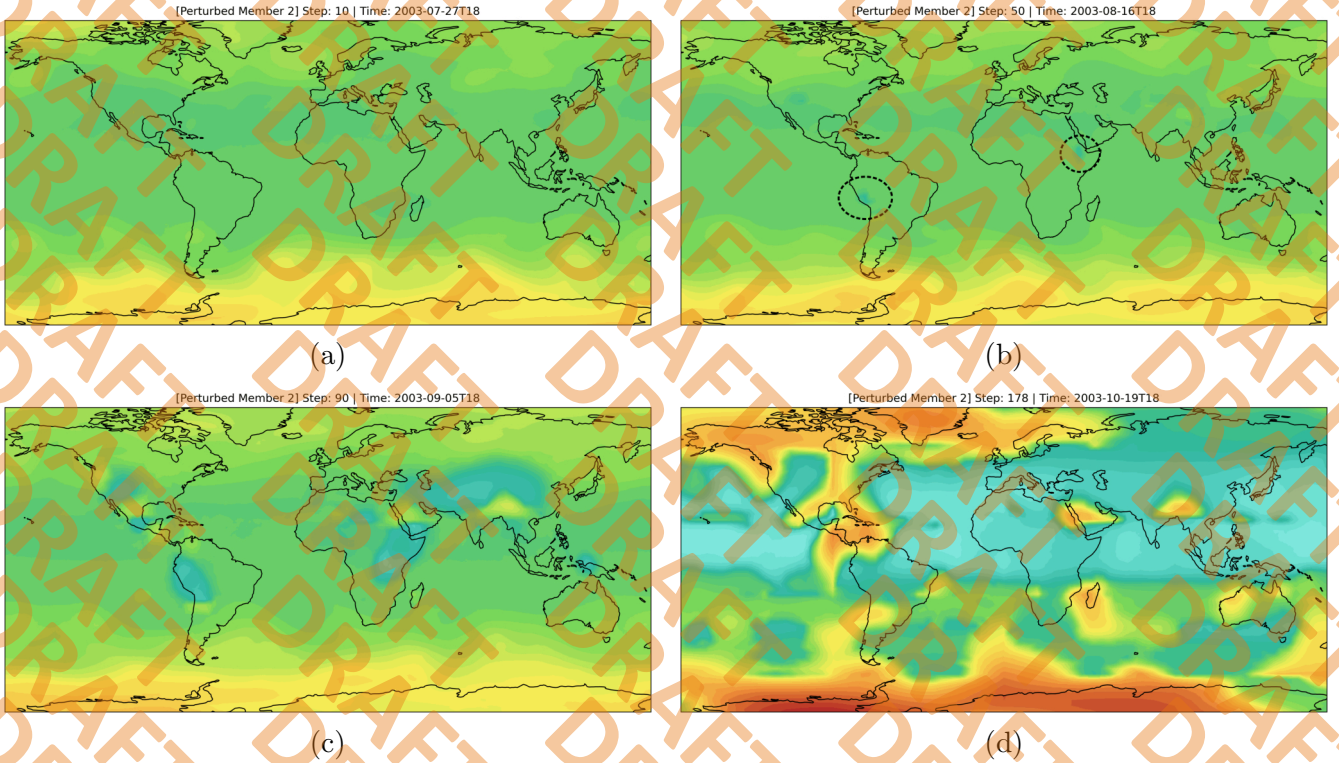


Figure S9 : Geopotential maps for different rollout time from a single run of AIFS-CRPS starting in July 2003, 12 days before the heatwave. Only two steps were saved per day in this run (12UTC and 18UTC) so 180 steps correspond to a summer of 90 days. (a) After 5 days, the forecast is normal. (b) after 25 days some artefacts start to appear in high altitude areas (circled). (c) After 45 days the artefacts have grown into large anomalies. (d) After 90 days, the artefacts have completely overtaken the forecast and made it unphysical and unusable.

height instead of temperature to compare with literature we get values of around 0.5 days^{-1} . This is just slightly out of the range usually found for the northern hemisphere that has an upper bound of 0.45 days^{-1} . Given that growth rates/Lyapunov exponents are dependent on the model resolution and quality overall it is not something that should be inherently bad.

Another interesting observation we make from Figure S10 is that some trajectories were able to reach temperatures as high or even higher than the historical reanalysis. These results are summarised in Table 2. This means we were able to recreate trajectories leading to possibly new unseen extreme heatwaves and justifies the use of this model for rare event algorithms.

D-b-3 Typicality and physics of the heatwaves generated

My goal here is to check whether the usual patterns observed for heatwaves (Section 2.5.1) are also the ones causing extreme heats in the runs that achieved as high or higher temperatures. For that we will be using composite maps. In climatology and meteorology, composite maps are averaged representation of multiple climate maps corresponding to specific periods or similar events. They help identify typical conditions associated to some phenomenon (heatwaves here). They can help synthesising a lot of plots in one map, but can suppress the diversity of individual cases.

For our three test cases of August 2003, July 2019 and June 2025, Figure S12 show results coherent with what is documented for heatwaves over western Europe (Kautz et al., 2022) and are therefore quite promising in the context of rare events analysis.



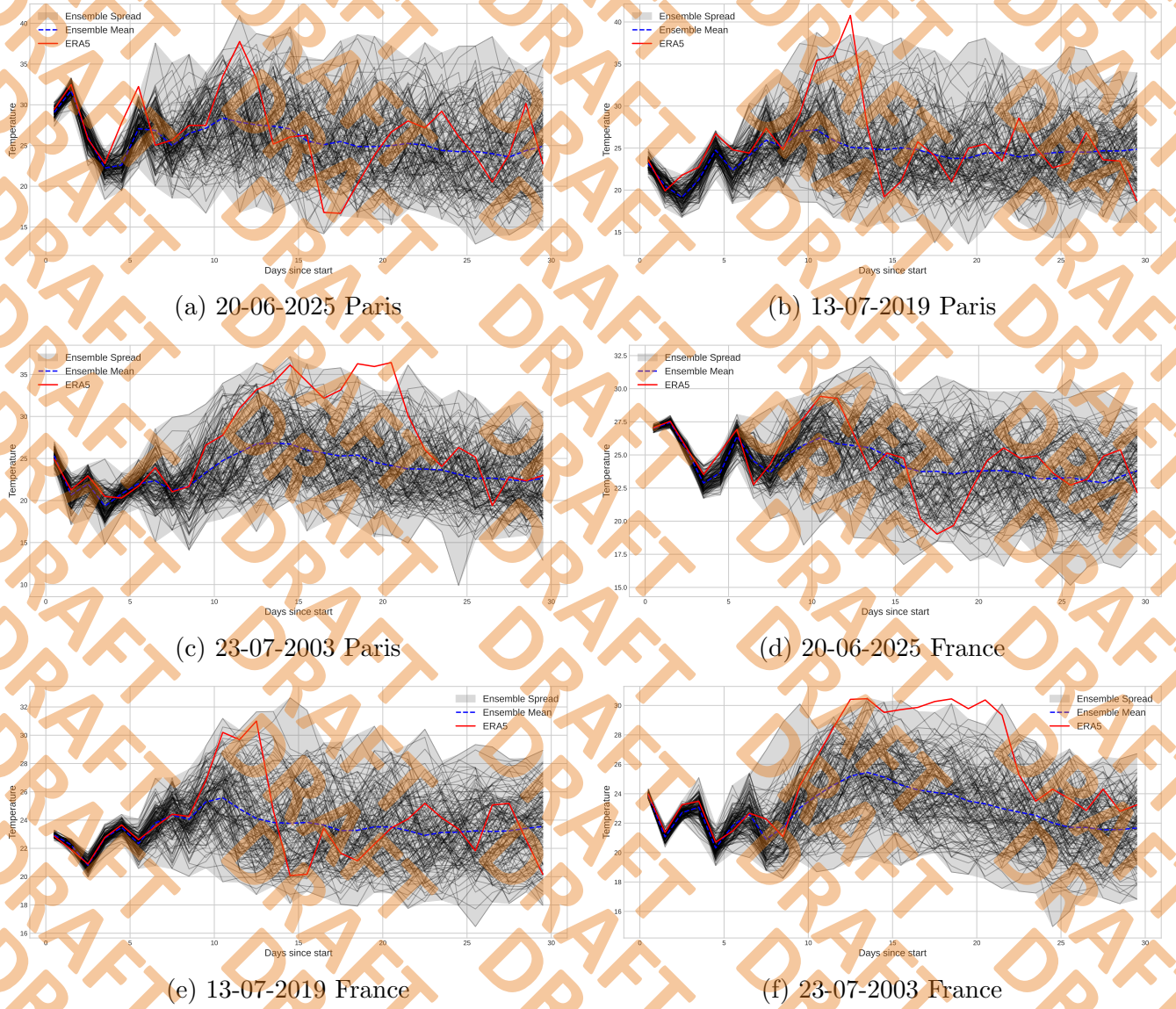


Figure S10 : Time series of temperatures for the Paris gridpoint generated using AIFS-CRPS from initial conditions 12 days before the (a) June 2025, (b) July 2019 and (c) August 2003 heatwaves. We notice that after only a few days the ensemble spread is large and that some trajectories reach temperatures as high or higher than the reanalysis.

There are also thermodynamic variables that are of interest, namely the MSE (Section 2.5.2) (Hotz et al., 2024; Noyelle et al., 2023b; Zhang & Boos, 2023) to check for convective instability in heatwaves generated. As expected, in the runs generated, when the temperature peaks, there is convective instability ($MSE_s \geq MSE_{500}^*$). This cools the surface through reduced insolation and precipitation. Convective instability is a behaviour that was to be expected but is still relevant to look for, AI models do not have equations or parametrisations that trigger this when it should. It is actually rather impressive that AIFS-CRPS is able to reproduce the convective instability behaviour of the system just from data.

D-b-4 Ability to generate long lasting and/or extremely hot heatwaves

One key aspect of rare events algorithms is that the model we are using, here AIFS-CRPS, has to be able to generate either extremely hot or long lasting (or both) heatwaves.

Looking at the 10 highest temperatures of the hottest simulated member against the 10 highest temperatures of that same period in ERA5, some preliminary conclusions can be drawn.



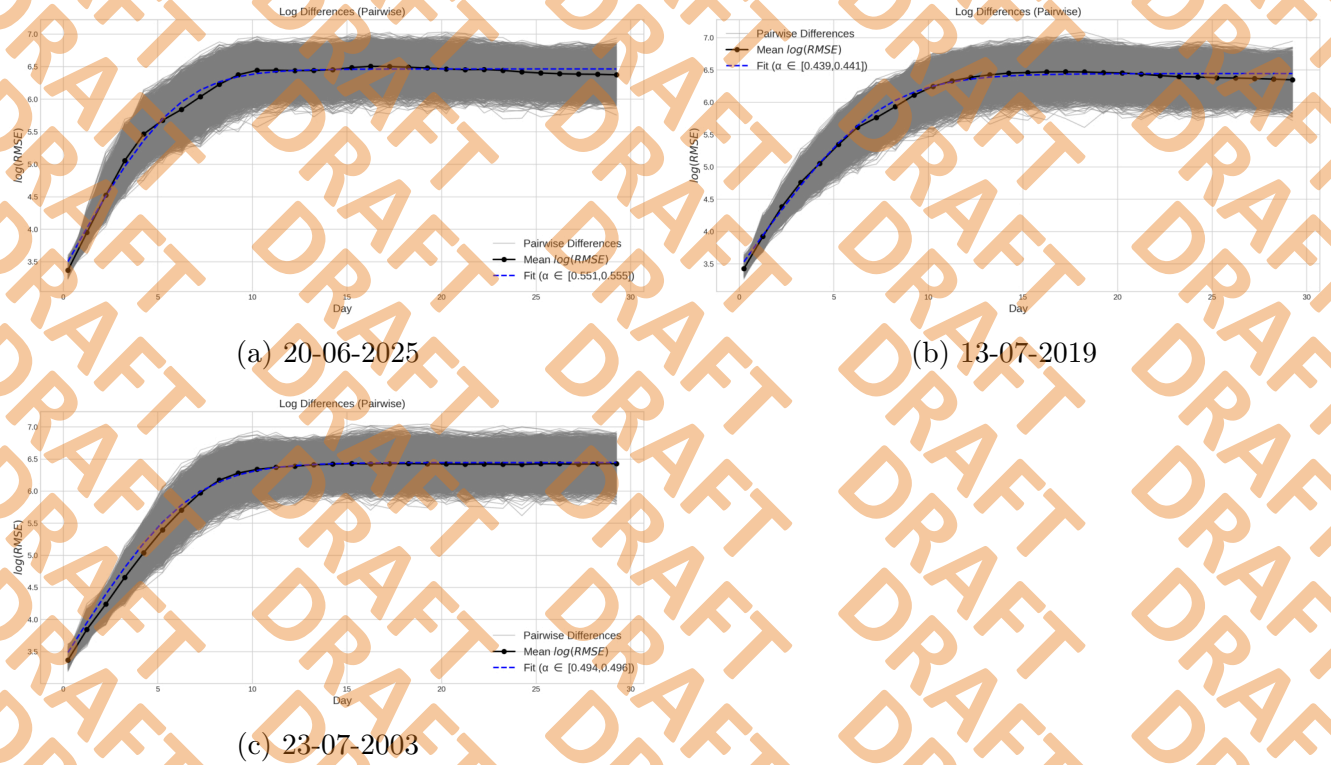
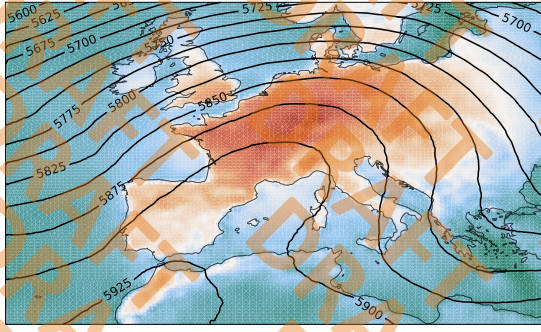


Figure S11 : Plot of all the pairwise log differences between (a) 100 and (b) 65 ensemble trajectories starting from initial conditions 12 days before the (a) June 2025, (b) July 2019 and (c) August 2003 heatwaves computed on the northern hemisphere surface temperature. The curves exhibit the expected behaviour of a linear growth followed by a plateau.

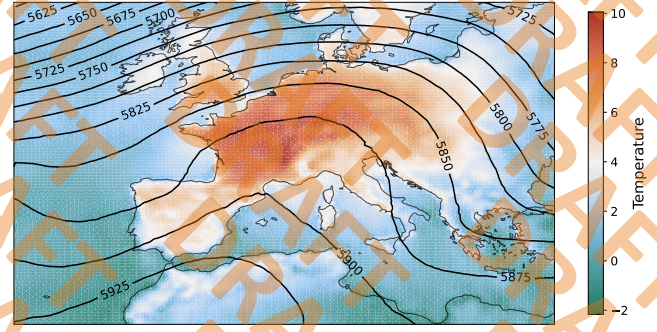
For long lasting heatwaves, Figure 14 and Figure 15 show that AIFS-CRPS seems to struggle to keep a high temperature over multiple days. When compared to ERA5, the 10 highest temperatures from the simulated member are – even though some are higher – less stacked in the top part of the graph. This would seem to indicate our model is unable to generate long lasting heatwaves. The same conclusions were drawn from analysing the other members. However, this is only a preliminary conclusion because the use of a proper rare event algorithm could compensate for that issue. For example, something as simple

Table S2 : Information on the boosted trajectories showed in Figure S10.

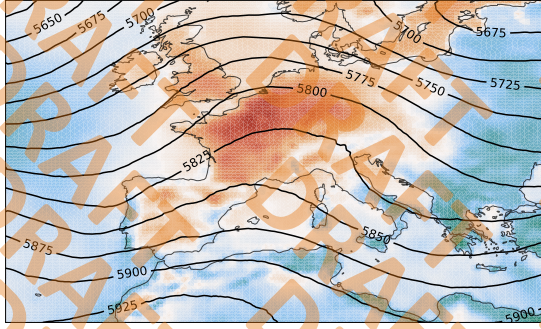
Location	Period (30 days)	ERA5 Max	Traj. Max	Traj. > ERA5 Max	Traj. > 2019 Record
Grid Point Closest to Paris	from 23/07/2003	36.42 °C	37.12 °C	3/129 (2.33%)	0/129 (0.00%)
	from 13/07/2019	40.78 °C	38.75 °C	0/129 (0.00%)	0/129 (0.00%)
	from 20/06/2025	37.77 °C	40.98 °C	10/129 (7.75%)	1/129 (0.78%)
Averaged over France	from 23/07/2003	30.46 °C	30.52 °C	1/129 (0.78%)	0/129 (0.00%)
	from 13/07/2019	31.00 °C	32.64 °C	3/129 (2.33%)	3/129 (2.33%)
	from 20/06/2025	29.42 °C	32.42 °C	28/129 (21.71%)	5/129 (3.88%)



(a) June 2025



(b) July 2019



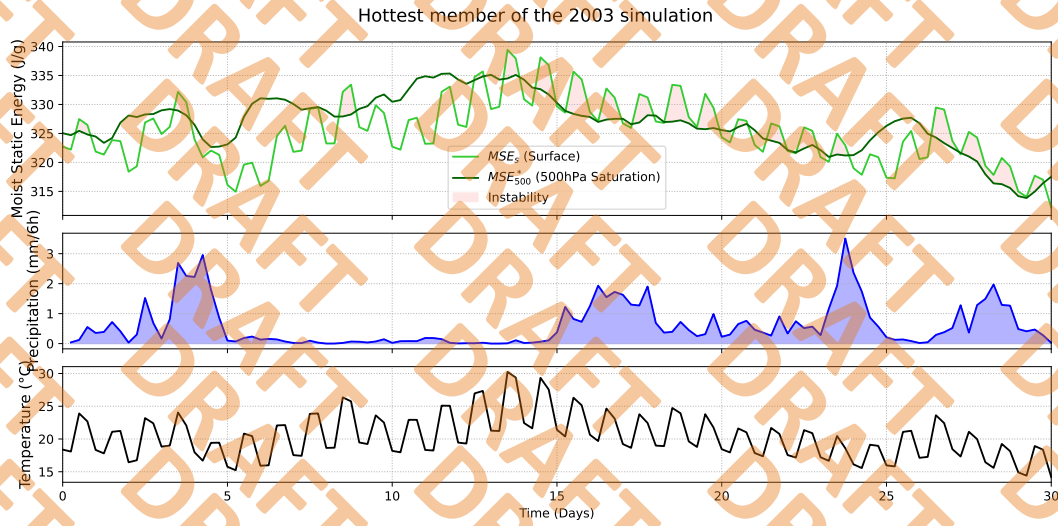
(c) August 2003

Figure S12 : Composite maps of temperature over western Europe for the 5 timesteps in all the runs where the temperature is the hottest at the Paris gridpoint. Temperatures are represented by colours and black contours are the geopotential heights. For the three test cases we observe an atmospheric blocking in the shape of an Ω

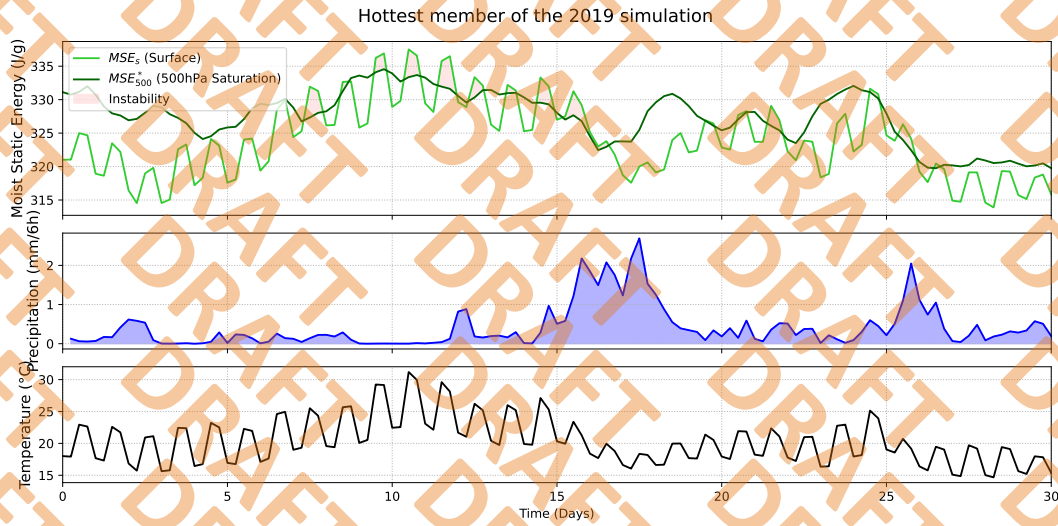
as a second ensemble boosting as performed in (Sippel et al., 2024) could be sufficient to maintain those temperatures in some members. An explanation could be that the model underestimates the persistence of Ω blocking because in the training data, only one heatwave of this type happened.

For record temperature peaks, we observe on the 2019 and 2025 heatwaves in Figure 14 and Figure 15 that the highest temperatures on both a single grid point or a spatial average can reach and even surpass those of ERA5. Surprisingly though, when looking at gridpoint temperatures, the simulation that yields the highest temperature is the one for June 2025. Which is interesting given one could have rightfully expected the July 2019 simulation to reach the highest temperatures since it actually did in ERA5. We suspect this is linked to the fact that July 2019 is part of the training data for AIFS-CRPS, or that 128 members just was not enough.

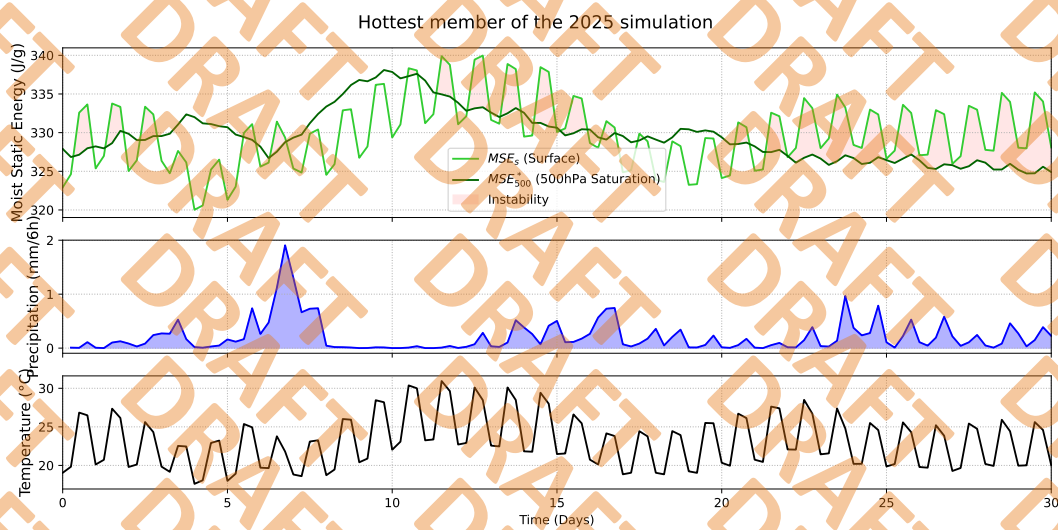




(a) August 2003



(b) July 2019



(c) June 2025

Figure S13 : Atmospheric profiles for the 2003 (a), 2019 (b) and 2025 (c) heatwaves.



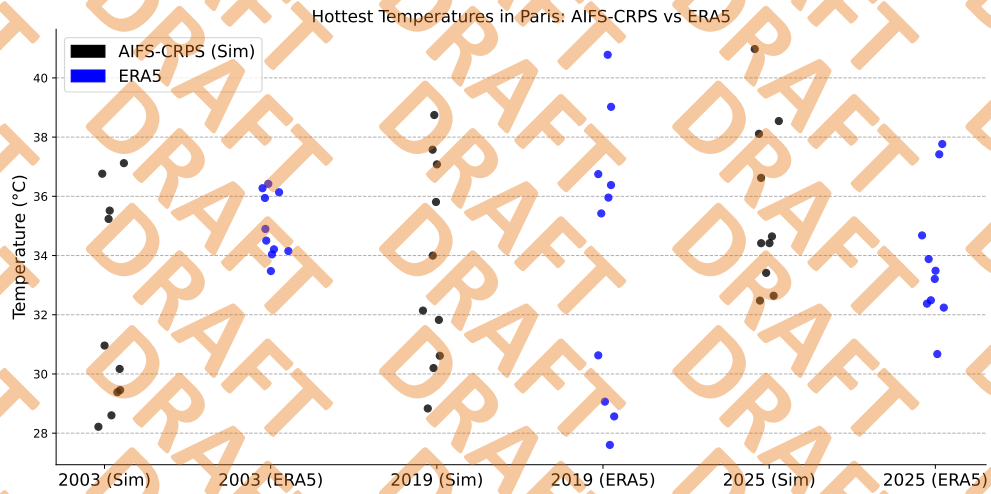


Figure S14 : Temperatures at 12UTC in Paris for our three heatwaves in ERA5 (blue) against the simulated member that reached the highest temperature (black).

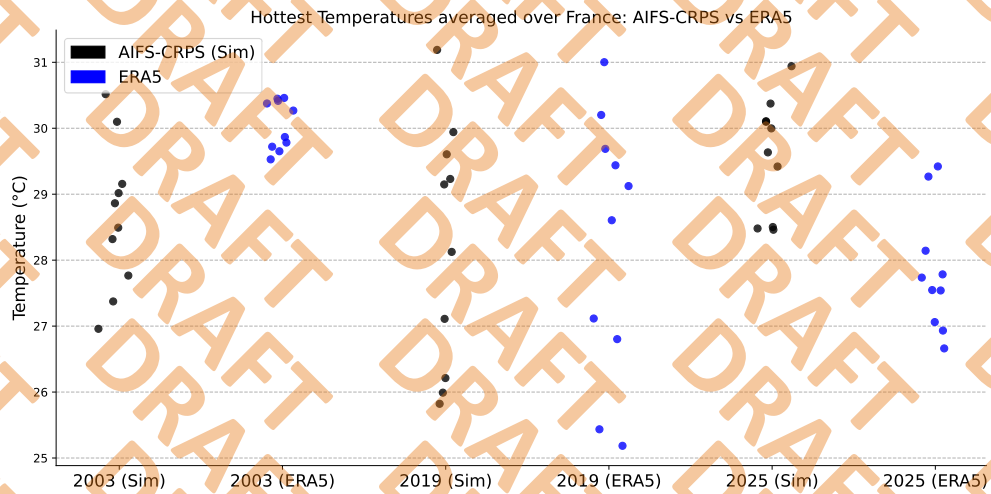


Figure S15 : Temperatures at 12UTC averaged over France for our three heatwaves in ERA5 (blue) against the simulated member that reached the highest temperature (black).



E COMPLEMENTS ON ITERATIVE BOOSTING

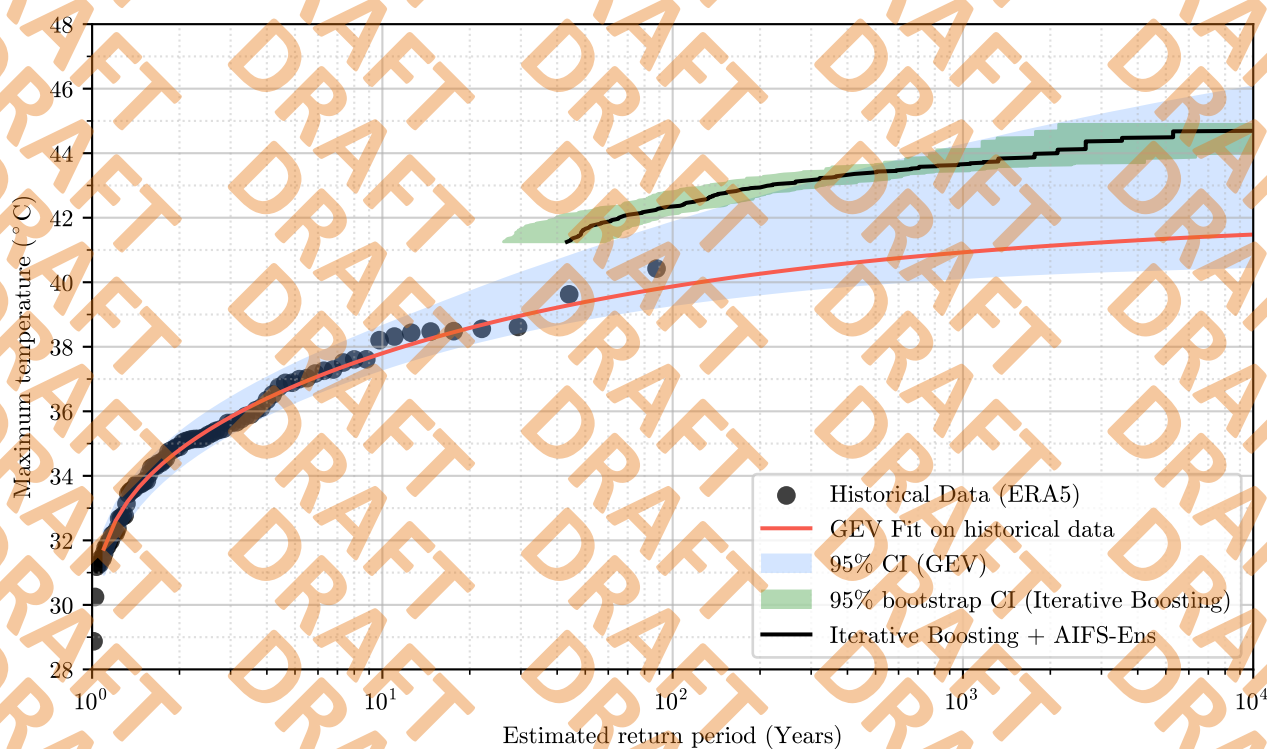


Figure S16 : Estimated return periods with the boosting estimator using the R1 trajectories and a GEV.



Title : Rare event algorithms using AI based weather forecasting.

Author : Paolo Chevalier

Keywords : climate extremes, heatwaves, rare event algorithms, AI weather models

AI based weather forecasting is becoming the new paradigm for fast and accurate short and medium range forecasting. In the meantime, the occurrence and intensity of extreme events is growing. In order to study them, we need to bias the distribution towards extreme events. This is done with rare event algorithms which up to now have been used only in conjunction with physical models. In this report, after having validated the physical and chaotic behaviour of ECMWF's AIFS-Ens model, I leverage its speed and performances to perform rare event algorithms, including a case study of the May 2026 heatwave, where I show that this heatwave was intensified by a cut-off low over the Iberian Peninsula. Finally, I present and evaluate a new method for rare event algorithm, Iterative Boosting, which is an extension of the regular Ensemble Boosting that allows us to reach extreme temperatures of up to 5°C over the historical maxima.

Titre : Algorithmes d'événements rares à l'aide de modèles météorologiques par IA.

Auteur : Paolo Chevalier

Mots Clés : extrêmes climatiques, vagues de chaleur, algorithmes d'événements rares, modèles météo par IA

La prévision météo par IA devient le nouveau paradigme pour des prévisions rapides et précises à court ou moyen terme. Dans un même temps, l'occurrence et l'intensité des événements extrêmes climatiques augmentent. Afin de les étudier nous avons besoin de biaiser les distributions vers les événements rares. Cela s'effectue grâce à l'utilisation d'algorithmes d'événements rares qui jusqu'à maintenant n'ont été utilisés qu'en conjonction avec des modèles physiques. Dans ce rapport, après avoir validé le comportement physique et chaotique de AIFS-Ens de l'ECMWF, je tire avantage de sa rapidité et de ses performances pour les algorithmes d'événements rares. J'étudie notamment la vague de chaleur de Mai 2026, où je montre que son intensité a été amplifiée par une goutte froide sur la péninsule ibérique. Enfin, je présente et évalue un nouvel algorithme d'événement rares, le Boosting Iteratif, qui est une extension de l'Ensemble Boosting usuel qui permet d'atteindre des températures extrêmes de plus de 5°C au dessus du maximum historique avec peu de trajectoires.

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